The Twin Criterion: Six Exception Positions

Paper 6. A single open cell in a sector is a twin pair unless it sits at one of six named places

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Abstract

Main result (Theorem 1). In the sector $(M^2, (M+6)^2)$, after every line $p \le M$ has acted, **at most six open cells are not twin pairs**, and their positions are explicit quadratics in M. So one open cell gives a twin pair unless it sits at one of six named places.

The six are the strikes of the two lines born inside the sector, minus the strikes lost to the line 3. In a window widened k-fold the six become $2k(2k+1)$ (Corollary 1a). Two of the six carry perfect squares and can be dropped. Of the other four, at most one can be open at a time **provided the root pair** $(M+2, M+4)$ **is not itself a twin** (Corollary 1d). The condition is needed: at $M = 9$ the root pair $(11, 13)$ is a twin, and 209 and 221 are open together.

Over a full period at most 31 exceptions fit, and no finite set of lines lowers that ceiling (§2.3). The odd Bonferroni truncation is a proved lower bound on the open count C_M, exact from the order $\max_d m(d) = \Theta(\log M / \log\log M)$ (Proposition 1).

A census of all 166,666,665 sectors below 10^9 finds no violation of any statement here, and measures the sifting dimension of the exception positions as three. **No progress toward the twin-prime conjecture is claimed.**

Numbering. This paper is one part of a set that was written as a single document and is now published in parts. Each part numbers its own results from one and is self-contained: a reference of the form [Pn, Thm 1] means Theorem 1 of Paper n, and an unqualified "Theorem 1" always means this paper's own. Result numbers therefore differ from those of the earlier seven-document releases, where the whole set shared the numbering of the single document.

How to read the claims in this paper. Statements set as Theorems, Propositions and Corollaries are proved, and the proofs are given. Anything described as *measured* is a computation over a stated finite range and is labelled as such where it occurs.

Keywords: twin primes, exception positions, sieve cycles, cell coordinates, singular series.

MSC 2020: 11N35, 11N05, 11A41.

Summary of the results in this part

§2 — The twin problem inside the framework

Result	What it says	Section
The pivot	Inside (u^2, v^2) with every line up to u switched on, **a cell that survives *is* a twin pair** — no primality test is needed. Everything downstream rests on this.	§2.1
Theorem 1	At most **six** cells of a sector can be open without being twin pairs, with their positions given explicitly in M.	§2.2
Corollary 1	The dichotomy: if $(M{+}2, M{+}4)$ is not a twin then $\lvert S_M \cap E_M \rvert \le 2$, so the sufficient count falls from $C_M \ge 7$ to $C_M \ge 3$.	§2.2
Corollary 1a	The six generalise: in $(M^2, (M+6k)^2)$ with $M > (6k)^2$ there are $2k(2k{+}1)$ candidate positions and no open cell with two composite endpoints.	§2.2, App. A.3
Corollary 1b	A line above 31 closes at most one of the six exception cells, unless it divides $M{+}8$, when it closes two. The large lines cannot gather the exceptions.	§2.2, App. A.3
Corollary 1c	In the symmetric window between the squares of $6r \mp 1$, a single surviving cell of any kind gives a twin pair — the central cell by the primality of its own endpoints.	§2.2, App. A.3
Corollary 1d	If $(M{+}2, M{+}4)$ is not a twin, discarding cells with a square endpoint leaves at most **one** exception, and none at all when $M \equiv 21 \pmod{30}$. Without that condition the statement is false — see $M = 9$.	§2.2, App. A.3
Corollary 1e	The sectors carrying an open non-square exception have density zero, and in the quantitative form 1e' at most $O(X/\log^3 X)$ of them — the four positions are sifted in dimension three, measured over 1.7×10^8 sectors.	§2.2, App. A.3
Remark	Each of the six partner quadratics takes square-free values with at most two prime factors infinitely often, by Iwaniec and Lemke Oliver — separately, not simultaneously.	§2.2, App. A.3
Proposition 2	The admissible exceptional configurations in a block of $N = 35L$ sectors number $B_L \ll L/\log^2 L$, by the Selberg sieve.	§2.4, App. B.1
Theorem 2	A newly born line closes **at most one** twin cell in its own first window: $D_p \in \{0, 1\}$.	§2.5
Corollary 2	$D_p = 1$ exactly when two primality conditions hold together, so the error term is itself twin-like.	§2.5
Theorem 3	The bridge pair: $B' = (q - 2 - \chi_q(2))B$ — the only law here that knows the window is anchored at a square.	§2.6
Theorem 4	The summation identity over M consecutive square windows.	§2.6

Result	What it says	Section
Recorded, not used	Two constructions that do not help — the balanced window, and ownership through the cofactor — kept so they are not retried.	§2.8, App. B.3
Proposition 1	The odd Bonferroni truncation is a **proved** lower bound on C_M, evaluated in one pass by the multiplicity $m(d)$, and exact once the order reaches $\max_d m(d)$ — and that order is $\Theta(\log M/\log\log M)$ by Proposition 1', far below $\pi(M)$; the truncation error is exact at every odd order.	§2.9
Placement	The same finite-window obstruction, reached independently from the Goldbach side.	§2.10
Propositions 3–5	A mirror law for the sector index; the exact discrepancy $s(d-s)/(dH)$ of a Fejér window, whence $d/(4H)$; and $\ll d^3/(HT^2)$ for a moving one, with a sharp exponent. The kernels are classical and cited; together they gain a factor of nine on L_7 and **reach none of S_5 and beyond**.	§2.11, App. B.4

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1. Setting

We use Papers [P1] and [P2]–[P4] as follows. Papers [P9]–[P11] are companions, not sources: they are cited where their obstruction explains why a cut is placed where it is, and nothing else is taken from them.

- **[P1]** supplies the window combinatorics: the increments W_j of $\lfloor 2j^2/p \rfloor$, their uniform bound and their exact histogram.
- **[P2]** supplies the coordinates: the line $L_p(k) = p(p+2k)$ beginning at p^2, the grid L_3, the cells $C_b = (6b-1, 6b+1)$, and the square window with its $+8$ growth.
- **[P3]** supplies the exact cycle laws: the four-state law, its refinement by inheritance depth, arbitrary weights in $\Omega_{\leq z}$, the refinement by line size, and the disjoint ownership layers.
- **[P4]** studies the transfer to a window of length $\asymp z^2$. On the tested windows, linear depth weights measure 1.0000 to the reported precision, the tested soft truncations stay close to 1, and the sharp depth-zero indicator is the exceptional endpoint, with ratio near 0.80. These are measurements, not an exact window theorem.

The last point shapes this paper. The indicator of depth zero is the condition "both members of the cell survive", which is the twin condition. So, in the tested family, the largest transfer loss falls exactly on the quantity that becomes a primality/twin condition inside the moving square window.

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2. The twin problem inside the framework

2.1 When a surviving cell is automatically a twin pair

Let $u < v$ be consecutive integers coprime to 6. Every survivor of all lines $\leq u$ inside (u^2, v^2) is prime: a composite $x < v^2$ has a prime factor $\leq \sqrt{x} < v$, and no prime lies strictly between u and v.

Hence

$$\boxed{\text{a surviving } NN \text{ cell in } (u^2, v^2) = \text{a twin prime pair.}} \qquad (2.1)$$

The half of this about a single square gap — the least prime factor of an odd composite between n^2 and $(n+1)^2$ is at most n* — is Lemma 1A of [4]. What (2.1) adds is the passage to a cell: the surviving cell is a twin pair, not just a pair of survivors.*

What (2.1) does and does not give for a sector. Equation (2.1) is an equality, but it needs no prime strictly between u and v. The sector $(M^2, (M+6)^2)$ with $M \equiv 3 \pmod 6$ fails this: the lines M+2 and M+4 are born inside it, and their strikes are exactly the six positions Theorem 1 names. So for a sector the relation is (2.2):

$$C_M = T_M + X_M, \qquad 0 \le X_M \le 6, \qquad (2.2)$$

with X_M the number of open cells that are not twin pairs, all at the six named positions. **The sector's geometry can move the survivor count away from the twin count only through those six places, and by at most six.** An instance from the table below: at $M = 10{,}005$ the sector holds 504 open cells and 503 twin pairs. The extra one is the cell (100,180,079, 100,180,081), whose upper member is $(M+4)^2 = 10{,}009^2$ — position *C*. *Measured over the 49,999 sectors with $M < 3 \cdot 10^5$: $X_M = 0$ in 45,439 of them, 1 in 4,369, 2 in 186 and 3 in five, and never more — so the bound of six is not attained on that range.*

Since the discrepancy is bounded by an explicit constant, C_M is the twin count up to six, so it should track the Hardy–Littlewood prediction (a conjecture) for a window of $|W| = 12M + 36$ integers at height M^2. The comparison below is thus a diagnostic on the twin count itself, up to that $O(1)$. Write

$$\mathrm{HL} = \frac{2C_2\ |W|}{\log^2(M^2)}, \qquad \mathrm{SP} = |W_{\text{cells}}| \prod_{5 \le p \le M} \left(1 - \frac{2}{p}\right)$$

for that prediction and for the raw sieve product over the cells. Then:

M	cells	C_M	HL	C_M/HL	SP	C_M/SP
1,005	2,016	87	83.6	1.041	104.7	0.831
5,001	10,008	281	273.2	1.028	343.4	0.818
10,005	20,016	504	467.3	1.079	587.8	0.857
20,001	40,008	796	807.9	0.985	1,017.6	0.782
50,001	100,008	1,679	1,691.9	0.992	2,131.8	0.788
100,005	200,016	2,936	2,988.6	0.982	3,766.1	0.780

The same comparison has been made independently, in the same coordinates. Morpurgo [10] studies twin pairs between $(p_n-2)^2$ and p_n^2, writes the candidates as multiples of 6, discards those congruent to ±1 modulo each sieving prime — the classes of [P2, Thm 3] in additive form — and compares the count with $\prod(p-2)/p$ corrected by $(2e^{-\gamma})^2$, reaching agreement better than one per cent for p_n above $6\cdot10^6$. That paper is a prediction and a measurement, not a set of theorems, and none of the statements below appears in it. We cite it because it uses the same coordinates and the same comparison, reached independently, and because its agreement over a far longer range than ours supports the reading of (2.1) given here.

The first ratio approaches 1 from above and settles just below it, within about 2% from $M = 20{,}001$ on; the one outlier is 1.079 at $M = 10{,}005$. The second settles at 0.78. That is the classical gap between the sieve product and the truth at depth $\sqrt{x}$, not a property of these coordinates. **The point of the table is the first column pair, and it is a constraint rather than a confirmation.** It leaves no room for an argument that hopes to push C_M above the Hardy–Littlewood count through structure in the sector — its anchoring at a square, the coupling between the lines' phases, the path of the sector start. By (2.2) C_M differs from the twin count by at most six, so any such excess would have to be an excess in the twin count itself.

Two controls make the same point from the other side; each closed a proposed route. First, replacing the true phases by random ones, with the same lines and the same two classes per line, raises the survivor count from 1,679 to $2{,}131\pm34$ at $M = 50{,}001$. But that gap comes entirely from the height of the window, not its arithmetic: the same sieve on a block of the same length at a comparable height $y \in (M^2, 3M^2)$ gives $1{,}797\pm60$, within two standard deviations of the truth. Second, a control that *keeps* the phase coupling of the sector path — all lines in one layer take their phase from a single number — but changes that number gives $2{,}127\pm34$, indistinguishable from the fully independent $2{,}131\pm34$. **The coupling between the lines' phases is a description of the sector, not a constraint on it.**

2.2 Theorem 1: only six cells in a sector can be open without being a twin

Take the sector in the form used throughout this section: $M \equiv 3 \pmod 6$ and the interval $(M^2, (M+6)^2)$, which carries $A_M = 2M+6$ cells

$$C_j = \big(M^2+6j+2,\ M^2+6j+4\big), \qquad j = 0, 1, \ldots, 2M+5.$$

Theorem 1. Switch on every line $p \le M$. Then at most **six** cells of the sector can be open without being a twin pair, and their positions are given explicitly by $E_M = \underbrace{\left\{\tfrac{2M}{3},\ \ M+1,\ \ \tfrac{5M}{3}+2,\ \ 2M+3\right\}}_{\text{present only if } M+2 \text{ is prime}} \cup \underbrace{\left\{\tfrac{4M}{3}+2,\ \ 2M+5\right\}}_{\text{present only if } M+4 \text{ is prime}}$. Consequently **every open cell whose index lies outside E_M is a twin pair.**

Proof. The argument has three steps: identify the only possible least prime factor, list its products inside the sector, and place them.

Step 1: the least prime factor is M+2 or M+4. Let N be a composite endpoint of an open cell. Every prime factor of N exceeds M, since otherwise some line $p \le M$ would have closed it. The least prime factor cannot be $\ge M+6$ either, since then

$$N \ge (M+6)^2,$$

which is outside the sector. As $M \equiv 3 \pmod 6$ and N is coprime to 6, the only remaining candidates are M+2 and M+4.

Step 2: six products, not more. Take $p = M$+2 first. Its strikes below $(M+6)^2 = (p+4)^2$ are

$$p^2, \qquad p(p+2), \qquad p(p+4), \qquad p(p+6), \qquad p(p+8),$$

since the next one, $p(p$+10), already exceeds $(p+4)^2$. Of these,

$$p(p+4) = (M+2)(M+6)$$

is divisible by 3 and so lies on L_3, not in a cell; **four remain**. Now take $q = M$+4. Its strikes below $(q+2)^2$ are

$$q^2, \qquad q(q+2), \qquad q(q+4),$$

of which

$$q(q+2) = (M+4)(M+6)$$

lies on L_3; **two remain**. Six in total.

Step 3: their positions. Write each surviving product in the form M^2+6j+2 or M^2+6j+4. For instance

$$(M+2)^2 = M^2+4M+4 = M^2+6j+4, \qquad j = \tfrac{2M}{3},$$
$$(M+2)(M+8) = M^2+10M+16 = M^2+6j+4, \qquad j = \tfrac{5M}{3}+2,$$

and each j is an integer because $3 \mid M$. The other four are placed the same way.

Finally, if M+2 is composite then it has a prime factor $\le M$, so its four products were already closed by an older line and are not exceptions; likewise for M+4. ■

Verification. Zero violations over every $M = 9, 15, \ldots, 19{,}999$ — **3,332 sectors.** In each, every open cell outside E_M was checked to be a twin.

Example, $M = 9$. The sector (81,225) has 24 cells; $E_9 = \{6, 10, 14, 17, 21, 23\}$, carrying

$$121 = 11^2, \quad 143 = 11 \cdot 13, \quad 169 = 13^2, \quad 187 = 11 \cdot 17, \quad 209 = 11 \cdot 19, \quad 221 = 13 \cdot 17.$$

No other composite can sit in an open cell of that sector. Of the six, **three are actually open**: 169 at position C, and 209 and 221 at E and F. The sector has $C_9 = 10$ open cells and $T_9 = 7$ twins, so $T_M \ge C_M - 2$ fails here. This sector shows why Corollary 1d carries its hypothesis: the root pair (11,13) is a twin. It is also the smallest case, and so easy to read past.

The candidate set E_M has 0, 2, 4 or 6 elements, according to the primality of M+2 and M+4:

M+2	M+4	$\lvert E_M\rvert$
composite	composite	0
composite	prime	2
prime	composite	4
prime	prime	6

At $M = 999$ both $1001 = 7 \cdot 11 \cdot 13$ and $1003 = 17 \cdot 59$ are composite, so E_M is empty and **all 93 open cells of that sector are twin pairs** — as measured.

Two remarks on the sharpness of this count. E_M is a set of candidate **positions**; fewer cells are actually open without being twins. Position B carries $(M{+}2)(M{+}4)$ and so needs **both** M+2 and M+4 prime. When M+4 is composite its least prime factor is at most $\sqrt{M+4} < M$, so an older line has already closed that cell and it cannot be an exception. Hence the row "M+2 prime, M+4 composite" admits at most **three** actual exceptions, not four. And the caps are not reached on any tested range: over every $M \equiv 3 \pmod 6$ below 4000, the largest number of open non-twin cells in a sector is 0, 2, 2 and 3 in the four rows. What Theorem 1 asserts, and all that is used below, is the inclusion $S_M \setminus E_M \subseteq \{\text{twins}\}$.

Why six: the positions are the steps of the newborn lines. A line L_p is dormant from its first contact with L_3 at $3p$ until its birth at p^2. That stretch is $p(p\text{-}3)$, i.e. $(p-3)/2$ steps of $2p$, each owned by a smaller line. After birth it walks on in steps of $2p$. Because $p^2 \equiv 1 \pmod 6$, the walk lands in turn on an upper member, on L_3, and on a lower member when $p \equiv 1 \pmod 6$, and on an upper member, a lower member and L_3 when $p \equiv 5 \pmod 6$: either way one step in three falls on L_3 and strikes no cell. Only two lines are born inside the sector, M+2 and M+4. From its birth, L_{M+2} makes five steps before $(M+6)^2$ ($k = 0, \ldots, 4$, since $(M+2)^2 + 2(M+2)k < (M+6)^2$ forces $k \le 4$); $k = 2$ falls on L_3, and the other four are A, B, D, E. L_{M+4} makes three steps ($k \le 2$); $k = 1$ falls on L_3, and the other two are C, F. Four and two make six. This is Theorem 1 again, proved by counting steps instead of factor pairs, and it explains the number as well as proving it.

The same walk gives the ladder of Corollary 1a below without algebra. In $(M^2, (M+6k)^2)$ the newborn lines are $M{+}a$ for the $2k$ admissible offsets a. Walking each from its birth and dropping the steps on L_3 leaves $4k, 4k-2, \ldots, 2$ cells — the rows of that corollary — with $2k^2$ steps lost to L_3 in all and $2k(2k{+}1)$ cells kept. Checked against the corollary's list for $k \le 6$ on four windows each, twenty-four in all, with exact agreement. At $k = 2$ the four newborn lines make 11, 9, 5, 3 steps inside the window and lose 3, 3, 1, 1 of them to L_3.

What this buys, stated precisely. With C_M the number of open cells, $T_M \ge C_M - |E_M| \ge C_M - 6$, so a twin follows from $C_M \ge 7$. Theorem 1 replaces that by the much weaker requirement

$$S_M \not\subseteq E_M, \qquad (2.3)$$

where S_M is the set of open indices: **a single open cell suffices, provided it is not at one of six named places.** The six places are fixed by the geometry of the square, not chosen by the lines, whose phases are periodic and unrelated to M.

The lines 5 and 7 alone cut six to three, and the pair is not interchangeable. This uses both lines, so it is stated for $M \ge 9$; the sector $M = 3$ has neither line switched on and is excluded throughout. With $M = 6n{+}3$, the six positions are exact quadratics in n as absolute cell indices:

$$A = 6n^2 + 10n + 4, \qquad B = 6n^2 + 12n + 6, \qquad C = 6n^2 + 14n + 8,$$

$$D = 6n^2 + 16n + 9, \qquad E = 6n^2 + 18n + 11, \qquad F = 6n^2 + 18n + 13.$$

(verified for $n = 1, \ldots, 3999$ against the definition). Each is thus a function of n mod p modulo any p, so the state of all six under the lines 5 and 7 depends on n mod 35 alone. **The 35 classes are not a sample but the whole question**, and checking them is a proof, not an experiment. The check gives $\max |S_M \cap E_M| \leq 3$ after those two lines alone, with only four maximal patterns —

$$\{A, B, C\}, \quad \{A, E, F\}, \quad \{C, D, F\}, \quad \{C, E, F\},$$

with $A, \ldots, F$ the six positions in the order listed above. The collapse from $2^6 = 64$ conceivable patterns to four makes the next step possible.

The pair is special. Line 5 alone leaves four open; adding 7 gives three; **adding 11, then 13, changes nothing** — same bound, same four patterns. The reason is structural: a larger line can take a phase that strikes none of the six, and the Chinese remainder theorem combines it freely with the phase of 5 and 7 that leaves three. **So the route "add more small lines until the six are closed" is shut.** We record it so it is not retried in another notation.

> **Corollary 1 (the dichotomy).** Suppose (M+2, M+4) is not a twin pair. Then $|S_M \cap E_M| \leq 2$, and consequently $C_M \geq 3 \implies$ the sector contains a twin, or $(M + 2, M + 4)$ is one.

Proof. Theorem 1 splits the six positions by what each one needs:

$$A, D, E \text{ require } M + 2 \text{ prime}; \qquad C, F \text{ require } M + 4 \text{ prime};$$

$$B \text{ carries } (M + 2)(M + 4) \text{ and so requires both.}$$

If the open set met both groups, M+2 and M+4 would both be prime and (M+2, M+4) would be the twin. So under the hypothesis the open set lies inside $\{A, D, E\}$ or inside $\{C, F\}$, and B is closed.

Each of the four maximal patterns above meets both groups, so none survives. From the 35 classes, the patterns that do survive are

$$\varnothing, \quad \{A\}, \quad \{C\}, \quad \{F\}, \quad \{A, D\}, \quad \{A, E\}, \quad \{C, F\},$$

all of size at most two. ■

What that list is a list of. These are the patterns open **at the level of a residue class modulo 35**: the sets the lines 5 and 7 alone leave open that fit the dichotomy. The set actually open in a sector is a subset of one of them, since larger lines may close more. Over the 2,992 sectors among $M = 9, \ldots, 19{,}999$ with a non-twin root pair, the realised sets are these seven plus $\{D\}$ and $\{E\}$ alone, which arise when a larger line closes A. The bound $|S_M \cap E_M| \leq 2$ is unaffected.

Verification. Over the same 3,332 sectors $M = 9, 15, \ldots, 19{,}999$: the maximum of $|S_M \cap E_M|$ is 3 in the 340 sectors where (M+2,M+4) is a twin and **exactly 2 in the other 2,992**. The forcing is never violated (an open member of $\{A, D, E\}$ with M+2 composite, of $\{C, F\}$ with M+4 composite, or B open without the twin), and the open set never meets both groups without the twin. **The separation is exact: three open positions occur only when the twin is already there.** Both bounds are attained — three in 3 sectors, two in 13 of the twinless ones — so neither is an artefact of a short range.

Five residue classes in which the two lines close all six. Of the 35 classes, five leave nothing open: $n \equiv 0, 8, 23, 25, 33 \pmod{35}$, that is

$$M \equiv 3,\ 51,\ 141,\ 153,\ 201 \pmod{210}.$$

There $S_M \cap E_M = \varnothing$ from the lines 5 and 7 alone, so **every** open cell is a twin. Condition (2.3) stays the weakest form of the criterion, since it asks for no count. In these classes it is *equivalent* to the bare $C_M \geq 1$; in a general sector Corollary 1 lowers the sufficient count from $C_M \geq 7$ to $C_M \geq 3$. The gain is in the threshold, not the criterion. Over the 713 such sectors below $M = 30{,}000$: 444,958 open cells and **not one fails to be a twin**, against 155 non-twin survivors in the 428 ordinary sectors below $M = 3000$ alone. The restriction costs nothing in the count: the survivor density in these five classes, against all others, is 0.934 up to $M = 4000$, 0.997 up to 12000 and 1.000 up to 30000. Fixing the phases of 5 and 7 changes which cells die, not how many.

The six are the first rung of a ladder, and the ladder does not help. The argument does not need the window to stop at $(M+6)^2$. Widening it to $W_{M,k} = (M^2, (M+6k)^2)$, with the lines still at $p \leq M$, gives exactly $K_k = 2k(2k+1)$ exceptional positions — $6, 20, 42, 72, \ldots$ — and k=1 gives back E_M. But the ceiling grows like $4k^2$ and the open cells like $2kM/\log^2 M$, so **widening the window makes the margin worse** and $k = 1$ is optimal. Two more sharpenings hold at k=1. Cells with a perfect-square endpoint carry no twin, and there are at most two. With line 5 alone, at most one of B, D, E, F is open when the root pair is not a twin — none when $M \equiv 21 \pmod{30}$, only B when $M \equiv 27$. The exception census reaches $M = 10^9$. All of this is bookkeeping, not strength: $C^{\circ}_M \geq C_M - 2$, so it still amounts to Corollary 1.

Corollaries 1a to 1e, the census, and the remark on the partner quadratics are in Appendix A.3.

2.3 The exception budget over a full period, and the limit of local arguments

Corollary 1 bounds the exceptions in one sector. Consecutive sectors are not independent, and a whole period of them lowers the bound further, to a limit that can be found exactly.

The sectors tile: $(M^2, (M+6)^2)$ is followed at once by $((M+6)^2, (M+12)^2)$, so a prime forced near the top of one is forced near the bottom of the next. Take the 35 sectors of one period — the window $(M_0^2, (M_0+210)^2)$ with $M_0 = 210k + 3$. Write each exceptional position's need as offsets that must be prime (A needs M+2; C needs M+4; D needs M+2,M+8; E needs M+2,M+10; F needs M+4,M+8). The twinless hypothesis then forbids two forced offsets differing by 2 anywhere in the window, not just within a sector.

Summing the per-phase caps gives $5 \cdot 0 + 20 \cdot 1 + 10 \cdot 2 = 40$. The coupling costs three: three pairs of adjacent phases cannot both reach their caps, because the offsets clash across the join. For instance, a phase reaching two via $\{C, F\}$ forces M+8, and a successor reaching two the same way forces $M' + 4 = M + 10$. **A fourth adjacent pair with the same caps does not clash**, and that is the point: the obstruction lies in the offsets, not in adjacency. Hence 37. Adding the lines 11,13,17, an exhaustive scan over all $5 \cdot 7 \cdot 11 \cdot 13 \cdot 17/35 = 2431$ alignments gives

$$\sum_{i=0}^{34} |S_{M_i} \cap E_{M_i}| \leq 31 \qquad \text{(no twin in the window)},$$

with the ceiling 31 attained in 9 of the 2431 alignments and the 2431 values having mode 26.

That ceiling cannot be lowered by any finite set of lines, and this is a theorem, not a report of failed attempts.

What is and is not claimed. The budget is a maximum over residues. What is proved: for every finite set of lines, some choice of residues lets the extremal configuration survive them all, so no residue-based argument brings the ceiling below 31. It is *not* a claim that an integer M_0 realising the configuration exists, still less infinitely many. That would need all 59 polynomials prime at once, which is Hypothesis H and is not assumed here.

The system. The 9 extremal alignments carry 28 distinct maximal configurations. Each needs certain $M_0 + a$ prime — 27 to 30 of them — and the *partner* member of each of the 31 cells to survive. The partners are quadratics in n: $36n^2 + 60n + 23$ for A, $36n^2 + 84n + 47$ for C, $36n^2 + 96n + 53$ for D, $36n^2 + 108n + 67$ for E and $36n^2 + 108n + 79$ for F (verified against the definition for $n < 3000$).

The surviving configuration. Testing the full linear-and-quadratic systems for a fixed prime divisor kills **27 of the 28**. Only three primes kill: $q = 23$ kills 20, $q = 31$ kills 20, and $q = 19$ kills 3. The sets overlap: 13 configurations have one killer, 12 have two and 2 have three. *(A split of the 27 by "first" killer depends on the order primes are tried, so it is not reported; by least killer it is 3, 17 and 7.)* One survives, at $M_0 \equiv 448353 \pmod{510510}$, with 28 linear and 31 quadratic conditions: 59 polynomials of total degree 90. Every prime $q \leq 90$ leaves at least one admissible residue for t; the tightest are $q = 19$, 31 and 47, with exactly one each. For $q > 90$ a residue survives automatically, since polynomials of total degree 90 have at most 90 roots between them. Regenerated by `code/verify_exception_certificate.py`, which emits the 28 configurations, every prime that kills each, and the survivor's admissibility table.

The system is thus admissible at every prime, so by the Chinese remainder theorem the residues can be chosen at once against any finite list of further lines: no residue-based argument, however many primes it uses, reaches 30.

Nor does the birth order of the lines supply the missing constraint. One might hope that a line $M_0 + a$, forced prime by one exceptional cell, strikes another of the 31 once born. Modulo $p = M_0 + a$ we have $M_0 \equiv -a$, so a member $(M_0 + u)(M_0 + v) + e$ with $e \in \{0, \pm 2\}$ reduces to the integer $(u\text{-}a)(v\text{-}a)+e$. Here $|u|, |v|, |a| < 216$ while $p \asymp M_0$, so the line divides the member only if that small integer is zero. Excluding the cell's own factors, this needs $|u - a| \cdot |v - a| = 2$, hence $|u - v| \in \{1, 3\}$. **But u and v are the offsets $6i+d$ with $d \in \{2, 4, 8, 10\}$, so $|u - v|$ is even, and the collision is impossible.** The check finds none, for the 28 lines or any $M_0 + b$ with $b \leq 216$. The timing of the lines carries nothing the residues did not. The escapee survives exactly when its 59 irreducible polynomials are prime at once — the setting of Schinzel's Hypothesis H [12], which predicts infinitely many t realising it. **Proving the configuration impossible would mean contradicting that prediction for a specific family: not a weakening of the twin problem but an apparent strengthening.**

2.4 Blocks of consecutive periods, and why lengthening the block does not help

Section 2.3 fixes the ceiling at one period. A block of L consecutive periods raises the budget to B_L. One may hope the budget grows more slowly than the block, so that a long enough block forces a twin. It does not: **Proposition 2** gives the sieve dimensions of the five exception types and the order $B_L \ll L/\log^2 L$. That is the order of the twin count itself, so the two grow together and no block length separates them. The measured budgets $B_3 = 67$, $B_5 = 100$, $B_7 = 138$, $B_9 = 163$ fit that order.

The full account, with the block scan and the dimension computation, is in Appendix B.1.

2.5 Theorem 2: a new line kills at most one pair in its own first window

The line L_p is born in the sector $[p^2, (p+2)^2)$. Index gap-2 pairs by their offset from the centre $C = (p+1)^2$: $P(j) = (C + 2j - 1,\ C + 2j + 1)$ with $-p \le j \le p$, so the window holds exactly $2p{+}1$ pair slots. Let T_p^- count the surviving pairs before L_p is switched on, T_p^+ after, and $D_p = T_p^- - T_p^+$.

L_p has exactly three strikes in this window: p^2, $p(p{+}2)$ and $p(p{+}4)$, since $p(p+6) > (p+2)^2$.

Theorem 2. $D_p \in \{0, 1\}$ for every prime $p > 3$.

Proof. Three strikes, and each is disposed of by the grid L_3.

1. **p^2.** Of the two pairs it meets, only (p^2, p^2+2) lies in the window — the other has index -p-1. For $p > 3$, $p^2 \equiv 1 \pmod 3$, so $p^2 + 2 \equiv 0$: **that pair is already dead.**
2. **The strike divisible by 3.** One of $p{+}2$, $p{+}4$ is $\equiv 3 \pmod 6$, so one of $p(p{+}2)$, $p(p{+}4)$ lies on L_3 and was struck before L_p existed; both pairs it meets were already dead.
3. **The surviving strike.** Call it

$$H_p = \begin{cases} p(p+2), & p \equiv 5 \pmod 6, \\ p(p+4), & p \equiv 1 \pmod 6. \end{cases}$$

   In either case $H_p \equiv 5 \pmod 6$, so $H_p - 2 \equiv 3 \pmod 6$ and the pair $(H_p - 2, H_p)$ is **also already dead.**

Only $(H_p, H_p + 2)$ remains. ■

Corollary 2. $D_p = 1$ if and only if **two** primality conditions hold together: $p \equiv 5\ (6)$: $p{+}2$ prime and $p(p{+}2){+}2$ prime; $p \equiv 1\ (6)$: $p{+}4$ prime and $p(p{+}4){+}2$ prime.

Proof. The cell H_p survives the older lines exactly when its cofactor,

$$p + 2 \quad \text{or} \quad p + 4,$$

has no prime factor below p; being less than $2p$, it is then prime. And $H_p + 2 < (p+2)^2$ is not divisible by p, so if composite its least prime factor is below p and it was struck earlier. Hence $H_p + 2$ survives iff it is prime. ■

Verification. Corollary 2 predicts D_p from two primality tests; checked against the direct D_p for every prime $5 \le p < 4000$, with **zero mismatches**. Examples: $p{=}5$ gives 7 and 37, both prime, so $D_5 = 1$; $p{=}11$ gives 13 prime but $145 = 5 \cdot 29$, so $D_{11} = 0$; $p{=}13$ gives 17 and 223, both prime, so $D_{13} = 1$.

What this does and does not buy. Since $T_p^+ = T_p^- - D_p$ and every survivor in the window is prime (the cofactor argument of [P7, §2.2]), T_p^+ *is* the number of twin pairs in $[p^2, (p+2)^2)$. So Theorem 2 says:

T_p^- is the twin count of the window, up to an error of 0 or 1, and the error is characterised.

This is the strongest local statement in the paper, and it is not a reduction. T_p^- and T_p^+ differ by at most one, so proving anything about T_p^- is proving it about the twin count. In particular the sufficient condition "$T_p^- \geq 2$" is *stronger* than the conclusion "$T_p^+ \geq 1$", not weaker. Corollary 2 adds that even the gap between them is a twin-like coincidence — two primality conditions at once — so the error term is of the same nature as the quantity.

2.6 Theorem 3: the bridge pair, and the only law that knows about squares

Index gap-2 pairs by x on an absolute scale, the pair being $(2x{+}1,\ 2x{+}3)$. The window for odd m starts at $a_m = (m^2-1)/2$ and holds $2m{+}1$ slots, while $a_{m+2} - a_m = 2m+2$. **So consecutive square windows do not abut: exactly one pair slot falls between them,**

$$g_m = a_m + 2m + 1, \qquad \text{the pair } \big((m+2)^2 - 2,\ (m+2)^2\big),$$

whose upper member is the next square itself. The line splits as window, bridge, window, bridge, ..., with no slack and no overlap.

Let B count the bridges surviving a line set, over a full cycle of roots.

> **Theorem 3.** $B' = (q - 2 - \chi_q(2))B$, where $\chi_q(2)$ is the Legendre symbol: $+1$ for $q \equiv \pm 1 \pmod 8$ and -1 for $q \equiv \pm 3 \pmod 8$.

Proof. The bridge at root r dies under q when $q \mid r^2$, i.e. $r \equiv 0$ — one class — or when $r^2 \equiv 2 \pmod q$, which has $1 + \chi_q(2)$ solutions. The two conditions are disjoint for $q > 2$, so $2 + \chi_q(2)$ classes are lost. ■

Verification. $B = 2,\ 8,\ 32,\ 320,\ 3840,\ 53760,\ 967680$ for the line sets up to $3, 5, 7, 11, 13, 17, 19$, against $T = 1,\ 3,\ 15,\ 135,\ 1485,\ 22275,\ 378675$. Brute-forced from the definition for $\{3\}$, $\{3,5\}$, $\{3,5,7\}$, $\{3,5,7,11\}$: exact.

So the cycle fingerprint has four numbers, not three, with four distinct degrees:

$$(M, S, T, B) \longmapsto \big(qM,\ (q-1)S,\ (q-2)T,\ (q - 2 - \chi_q(2))B\big).$$

***B* is the only one of the four that knows the window is anchored at a square**, and the arithmetic that enters is whether 2 is a quadratic residue.

> **Theorem 4 (the summation identity).** Over M consecutive square windows,
> $$\sum_{i=0}^{M-1} T_{m+2i} = 2(m+M)T - B.$$

Proof. The M windows together with their M bridges tile a stretch of

$$a_{m+2M} - a_m = 2M(m+M)$$

pair slots: exactly $2(m{+}M)$ complete cycles, holding $2(m{+}M)\,T$ surviving pairs. The M bridge roots are M consecutive odd numbers; as M is odd they cover each class modulo M once, so exactly B survive. ■

Verification. Checked for $\{3,5\}$ and $\{3,5,7\}$ at $m = 9,\ 15,\ 101$ — six cases, all exact.

The collective bias, quantified. Against the density prediction T ($2m{+}2M{-}1$) for the same length, the windows hold exactly T-B fewer pairs; the measured sums of deviations are $-1, -5, -17, -185$

for the four line sets, matching T-B each time. Now

$$\frac{B}{T} = \prod_q \Big(1 - \frac{\chi_q(2)}{q-2}\Big),$$

which converges: measured 2.5554, 2.5517, 2.5614, 2.5615, 2.5622 at $z = 19,\ 10^3,\ 10^4,\ 10^5,\ 10^6$. The reduced product $\prod(1 - \chi_q(2)/q)$ reaches 1.604401 at $z = 10^6$ against

$$\frac{1}{L(1,\chi_8)} = 1.604556, \qquad L(1,\chi_8) = \frac{\log(1+\sqrt{2})}{\sqrt{2}} = 0.623225.$$

So the square geometry enters this framework through a Dirichlet *L*-value at 1 for the character modulo 8.

The bias is real but not usable: $T - B \approx -1.56T$ is a fixed number, independent of m, spread over M windows. The deficit per window is $O(T/M)$ and vanishes against the per-window average as M grows.

2.7 The square-phase mean, and the end of the "poor window" question

Is a window anchored at a square poorer than one placed anywhere? The answer is an identity, not an experiment: the mean over all square phases has the exact closed form (B.2.1), and on the tested range it tracks the generic density prediction within about 1% from $p = 29$ on. So anchoring at a square costs nothing in the mean, and the "poor window" question closes.

The derivation and the comparison table are in Appendix B.2.

2.8 Two constructions that do not help, recorded so they are not retried

Two natural constructions add nothing: a window balanced so every line strikes it equally often, and ownership read through the cofactor instead of the smallest factor. Both are exact and both reduce to statements already in the framework.

The two constructions and the measurements that close them are in Appendix B.3.

2.9 Proposition 1: a proved lower bound on C_M, and the order at which it becomes exact

So far every count of C_M here has been a direct enumeration. This subsection gives the first *proved* lower bound, from a classical inequality and the point where it terminates.

Fix a sector, with W its cells and $N = |W| = 2M + 6$. For each line $p \le M$ let $B_p \subseteq W$ be the cells it closes — two residue classes modulo p, by [P2, Thm 3]. For a cell d put

$$m(d) = |\{p \le M : d \in B_p\}|,$$

the number of lines striking it, and define

$$S_0 = N, \qquad S_i = \sum_{d \in W} \binom{m(d)}{i} \quad (i \ge 1). \qquad (2.4)$$

The right-hand side of (2.4) is the size of the i-fold intersections $|\bigcap_{p\in J} B_p|$ summed over all i-subsets J: each cell counts once per i-subset of the lines striking it. In this form the whole hierarchy takes a **single pass over the cells**, at cost $O(N \log\log M)$, instead of enumerating $\binom{\pi(M)}{i}$ intersections.

> **Proposition 1.** With the notation above, for every $\ell \geq 0$ $C_M \geq S_0 - S_1 + S_2 - \cdots - S_{2\ell+1}$, (2.5) and the alternating sum is **exactly** C_M as soon as the truncation order reaches $\max_d m(d)$.

Proof. A cell struck by exactly m lines contributes

$$\sum_{i=0}^{L} (-1)^i \binom{m}{i}$$

to the truncated sum. For $m = 0$ that is 1. For $m \geq 1$ and $L < m$ the partial alternating binomial sum equals $(-1)^L \binom{m-1}{L}$, which is ≤ 0 when L is odd. So every struck cell contributes at most 0 and every open cell exactly 1, giving (2.5). For $L \geq m \geq 1$ the sum is the complete alternating binomial sum $(1-1)^m = 0$. So once $L \geq \max_d m(d)$ every struck cell contributes 0 and every open cell 1, and the total is exactly C_M. ■

The inequality is the classical odd Bonferroni truncation. The multiplicity form (2.4), and its termination at $\max_d m(d)$, make it usable here. The same evaluation appears independently in Nguyen [11], in the Goldbach setting of symmetric pairs about a multiple of a primorial — see §2.10.

How far the low orders reach. Write L_j for the truncation at order j (this L is the truncation order, **not** the line L_p of [P2], which does not occur here). Measured:

M	cells	L_1	L_3	L_5	L_7	C_M
9	24	7	**10**	10	10	10
15	36	1	**11**	11	11	11
21	48	-13	**7**	7	7	7
51	108	-72	2	**10**	10	10
105	216	-208	-27	20	**21**	21
141	288	-313	-76	25	**29**	29
201	408	-501	-191	19	**28**	28
381	768	-1104	-492	8	**47**	47
501	1008	-1538	-839	-40	46	47
753	1512	-2490	-1431	-93	74	75

The first moment $L_1 = N - S_1$ is the union bound, and it dies at once: $\sum_p 2/p$ passes 1 at $M = 13$ and then grows like $2\log\log M$, so L_1 is negative from $M = 21$ on and carries no information. Order three is exact through $M = 21$. Order five is exact through $M = 51$; at $M = 105$ it gives 20 against $C_M = 21$, and at $M = 141$ it gives 25 against 29, so an earlier version's "through $M = 141$" contradicted the row beside it. Order seven is exact through $M = 381$ and short by one at $M = 501$ and at $M = 753$.

Where the hierarchy terminates, and it is not at the mean. By Proposition 1 the truncation is exact once the order reaches $\max_d m(d)$. The mean of $m(d)$ over the sector is $\sum_{p\leq M} 2/p \approx 2\log\log M$ — about 4.1 at $M = 10^6$. An earlier version inferred from that mean that

the **maximum** is also of order $\log\log M$. That inference is invalid — a maximum over $2M+6$ samples is not controlled by their mean — and the conclusion is false. Proposition 1' below gives the correct order. The measurements:

M	10^3	$5\cdot10^3$	10^4	$5\cdot10^4$	10^5	$2\cdot10^5$	$5\cdot10^5$	10^6
$\max_d m(d)$	9	9	10	11	11	12	13	13
least odd j with $L_j>0$	7	7	9	9	9	9	9	11

Across three orders of magnitude in M the order needed rises only from seven to eleven — against $\pi(M)=78{,}498$ lines at the top of that range. So reading "one would have to take every order" as "$\pi(M)$ orders" is wrong by four orders of magnitude, and the exact point is a quantity (2.4) already computes for free.

> **Proposition 1' (the exact termination order).** Let $p_1=5, p_2=7,\dots$ be the primes from 5 on, $Q_k=p_1\cdots p_k$, and $K(t)=\max\{k\ge0 : Q_k\le t\}$. Then for every $M\equiv3 \pmod 6$ with $M\ge9$, $K(2M+6) \le \max_d m(d) \le K\big((M+6)^4\big)$, and consequently $\max_d m(d)=\Theta\big(\log M/\log\log M\big)$.

Proof. **Lower bound.** Put $k=K(2M+6)$. Every prime dividing Q_k is at most M: for $k=1$ because $5\le M$, and for $k\ge2$ because $Q_k\ge5p_k$ gives $p_k\le(2M+6)/5\le M$. Since $\gcd(6,Q_k)=1$ the congruence $6d\equiv1 \pmod{Q_k}$ has one solution class, and the sector's $2M+6\ge Q_k$ consecutive indices contain it. That cell has $6d-1\equiv0$ modulo every $p_i\mid Q_k$, so all k lines strike it. Both its endpoints exceed $M^2\ge p_i^2$, so each line is born and the strike is active. The witness is explicit: $d=a+(6^{-1}-a \bmod Q_k)$. **Upper bound.** If m distinct primes strike a cell their product divides $(6d-1)(6d+1)<(M+6)^4$, and that product is at least Q_m. **Asymptotics.** $\log Q_k\sim k\log k$ by the prime number theorem, so $K(t)\sim\log t/\log\log t$, and the two bounds differ only in the constant, which lies between 1 and 4. ■

The finite data cannot decide this. Over the tabulated range the two candidate orders look the same: $\max_d m(d)$ divided by $2\log\log M$ reads $2.33, 2.10, 2.25, 2.31, 2.25, 2.40$ at $M=1005$ to 200,001, and divided by $\log M/\log\log M$ it reads $2.52, 2.26, 2.41, 2.42, 2.33, 2.46$ — both flat, neither favoured. So the table is no evidence against the earlier claim and never conflicted with it. The construction in Proposition 1', not the measurement, settles the order.

> **Proposition 1" (the exact truncation error).** For every odd $j\ge1$, $C_M-L_j=\displaystyle\sum_{d:m(d)>j}\binom{m(d)-1}{j}$, every summand strictly positive. Hence, for odd j, $L_j=C_M$ **if and only if** $j\ge\max_d m(d)$.

Proof. Cell by cell, from the partial alternating binomial sum $\sum_{i\le j}(-1)^i\binom{m}{i}=(-1)^j\binom{m-1}{j}$ used in the proof of Proposition 1. Summing over the cells gives the identity; positivity of each term gives the equivalence. ■

This makes the termination criterion of Proposition 1 necessary as well as sufficient, and gives the error at every truncation, not only its sign. With h_r the number of cells of multiplicity r, the remainder is $\sum_{r>j}h_r\binom{r-1}{j}$, so a certificate for the multiplicity histogram certifies the whole hierarchy at once.

Two orders that must not be confused. The least order at which the truncation is *exact* is $\max_d m(d)$, of order $\log M/\log\log M$. The least order at which it first turns *positive* — what a lower bound on C_M needs — is a different quantity, tabulated above as seven to eleven; nothing here fixes its growth. Proposition 1' bounds the first and says nothing about the second.

Regenerated by `code/verify_max_multiplicity.py`.

And what this does not buy. The bound is proved and the order is small; the *number of terms* is not. A theorem about all M from (2.5) needs asymptotic control of $S_1, \dots, S_j$ for $j \asymp \log\log M$, and S_j is a sum over j-tuples of primes of the same shape that makes S_1 diverge. So the hierarchy moves the difficulty from "the union bound is negative" to "a uniform estimate is needed for sums of order $\log\log M$" — a shorter distance of the same kind. It is recorded as an exact computational tool and a sharper statement of where the estimate is missing, not as a route.

2.10 The same obstruction from the Goldbach side

The finite-window difficulty this section keeps meeting — a full CRT period has positive density, but the interval of one centre is a *translated fragment* of it — was reached independently, in almost the same words, by Nguyen [11]. His setting is Goldbach, not twins: symmetric pairs $\{C-d, C+d\}$ about a centre C that is a multiple of a primorial. His pairs have fixed **sum** where ours have fixed **difference**, and his window is anchored at a multiple of a primorial where ours is anchored at a square. Under that translation the frameworks match term by term: the two forbidden cell classes $\pm 6^{-1} \pmod p$ of [P2, Thm 3] are his one or two forbidden lift residues; C_M matches his survivor sets; and the survivor density $\prod(1-2/q)$ is the same product.

One point in that dictionary needs care, and Nguyen raised it. His $U(C)$ is deliberately *conservative*: the offsets avoiding every obstruction congruence. The actual terminal set $T(C)$ also holds the **endpoint-prime exceptions** — offsets whose divisible endpoint is the prime q itself, so nothing is destroyed. In general $U(C) \subsetneq T(C)$. In his example at $C = 30$, the offset 23 gives the surviving pair $\{7, 53\}$, while 19 gives the truly destroyed endpoint $49 = 7^2$.

On our side the two coincide, because **the window sits above M^2 while every line is at most M**. An endpoint equal to a sieving prime is impossible: a member of a cell in $(M^2, (M+6)^2)$ exceeds $M^2 \geq p^2 > p$ for every $p \leq M$. So there are no endpoint-prime exceptions, and C_M equals both $|U|$ and $|T|$ in his letters. The equality comes from anchoring at a square, not from the definitions; a window placed elsewhere would have to tell them apart.

Two things are worth recording. First, the correspondence is evidence, not coincidence: two constructions built for different problems, with no contact, isolate the same difficulty and call it the central one. Second, the tools complement each other. The Bonferroni evaluation of §2.9 is taken from there; the measurement of $\max_d m(d)$ and the covering control of [P9, §3.5] were not made there and bear on its open problems. Nguyen has confirmed in correspondence that $S_j = 0$ for $j > \max_d m(d)$ in his notation too, so the effective depth of the inclusion–exclusion is that maximum, not the number of sieving primes. Whether it also grows like $\log\log$ on the primorial wheel is open. Nguyen claims no Goldbach theorem and no new infinite family, and says so repeatedly; the reference is to contemporaneous independent work, not a settled result.

2.11 Smoothing the window, and where it does not reach

A sharp window pays a boundary error of order 1 per residue class; a Fejér kernel removes most of it. Three exact statements are proved: a mirror law for the sector index (Proposition 3), an exact discrepancy formula for the Fejér window, with the classical bound $d/(4H)$ as corollary (Proposition 4), and a bound $\ll d^3/(HT^2)$ for a window that also moves (Proposition 5). The kernels, the order, the pointwise estimate and the $\csc^4$ sum are classical and cited as such; we have not found the exact discrepancy formula stated anywhere.

Smoothing gains a factor of about nine on L_7 at $M = 50{,}001$ and does not change its sign. **On our reading it does not reach the part that decides the answer:** the share of S_i carried by moduli below the window scale is 90%, 45%, 9%, 0.2% and 0% for $i = 1, \dots, 5$. So it controls the term already dead at $M = 21$ and none of the terms that fix the sign.

The three propositions with their proofs, the measured tables, and a further route closed by an implication toward Jacobsthal's function, are in Appendix B.4.

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3. What this paper establishes, and what it does not

Proved in this paper.

- The six exception positions (Theorem 1), the dichotomy that three of them can be open only when $(M{+}2, M{+}4)$ is itself a twin (Corollary 1), and the fact that no line above 31 closes two of them unless it divides $M{+}8$ (Corollary 1b).
- The ladder in the width of the window and the square-endpoint sharpening — Corollaries 1a, 1c, 1d, 1e, Appendix A.3.
- The exception budget of 31 over a full period, and the proof that no finite set of lines lowers it — §2.3.
- The block budgets $B_3 = 67$ and $B_5 = 100$, the sieve dimensions of the five exception types, and the resulting order $B_L \ll L/\log^2 L$ — Proposition 2, §2.4.
- The single-kill bound in a line's own first window and its characterisation — Theorem 2, Corollary 2.
- The bridge law and the summation identity over a cycle of windows — Theorems 3, 4.
- The odd Bonferroni lower bound on C_M, its multiplicity evaluation, and its termination at $\max_d m(d)$ — Proposition 1, §2.9.
- The mirror law for the sector index, the exact Fejér discrepancy formula, and the moving-window bound, with the certificate that smoothing preserves the target — Propositions 3, 4, 5, §2.11. The kernels, the pointwise estimate and the $\csc^4$ sum are classical and are cited as such.

What this paper imports, and it is only the coordinates. [P1] for the window combinatorics, [P2] for the lines, the cells and the square window, [P3] for the exact cycle laws, and [P4] for the transfer measurements — the four listed in §1. Nothing from [P5], [P7] or [P8] is used in any proof above; those are companions, with their results in their own places.

Measured, and labelled as such where they occur. Each item is a computation over a stated finite range and supports a claim about that range only.

- The agreement of C_M with the Hardy–Littlewood prediction — within about 2% from $M = 20{,}001$ on, with one row 7.9% out — and the two phase controls that go with it — §2.1.

- The block values $B_7 = 138$ and $B_9 = 163$, and the values quoted for $L = 21, 51, 101$ — §2.4. They were computed with pruning and are stable across twelve and eight consecutive primes respectively; **read them as lower bounds with strong stability, not as certified maxima.**
- The numerical agreement of the square-phase mean with the generic density — §2.7 — and the two constructions of §2.8.
- The growth of $\max_d m(d)$ and of the least sufficient Bonferroni order, tabulated in §2.9. The inequality there is proved; the growth rate is measured.
- The gains from smoothing, the share of each S_i below the threshold $d < r$, and the least admissible half-width against the largest gap, tabulated in §2.11. The bounds there are proved; the tables are measured.

Where the supporting material is. The counting details and the ladder in the width of the window are in Appendix A; A.3 carries Corollaries 1a to 1e and the census to $M = 10^9$. Appendix B holds four accounts moved from §2: the block budgets (B.1), the square-phase mean (B.2), the two constructions that do not help (B.3), and the smoothing bounds with the mass they miss (B.4). Each has a stub in §2 with its conclusion and status.

Covered by the verification scripts. `verify_exception_dichotomy.py` covers the closed forms of the six positions, the $\{5, 7\}$ table, Corollary 1, the whole of the period budget of §2.3 (the per-phase caps $5 \cdot 0 + 20 \cdot 1 + 10 \cdot 2 = 40$, the coupling to 37, the ceiling 31 over all 2431 alignments, the nine that attain it, and the partner quadratics), and Step 1 of Proposition 2 over all 423 primes below 3000. `verify_bonferroni_depth.py` covers Proposition 1 — the geometry of §§2.9 and [P2, §4.5], the exception positions of Theorem 1 over every sector below $M = 2500$, the exactness of the alternating sum at order $\max_d m(d)$, and the published values. `verify_exception_certificate.py` covers the 28 maximal configurations of §2.3, the primes that eliminate 27 of them, and the admissibility of the 59 polynomials of the survivor. `verify_max_multiplicity.py` covers the order $\Theta(\log M / \log\log M)$ of $\max_d m(d)$ in §2.9 and its explicit witness. `verify_new_additions.py` covers the exact phase set of [P2, Cor 1] over 428 primes, the diamond-centre corollary over 6,320 pairs, the coincidence gap of [P2, Cor 3], the third channel and pair count of [P3, Cor 1–2], and the two constructions of §2.8 — the rotation of the balanced word over 20 windows and the clean-owner theorem.

Not here. Every statement above is *exact* — an identity, a cap, or an explicit list. None gives a lower bound that stays positive for all large M — Proposition 1 is a lower bound, but its truncations turn negative — and such a bound is what the twin conjecture needs. Each criterion above needs a lower bound on a quantity that exceeds the twin count of a sector by at most a constant. So the criteria are exact reformulations, not routes.

The measurements behind that, the four tests the framework was checked against, and where and why it stops, are in **Papers 9 to 11**.

The computations and much of the prose in this paper were prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running the computations, and for auditing the papers against their own scripts. All statements were checked by the author, who is responsible for them; the repository README sets out the division of labour in full.

Appendix A — Counting details for §2

The two routine computations below are quoted in §2 and kept here so they do not interrupt §2.

A.1 Raw cell count and main term

$$C = \frac{v^2 - u^2}{6} - 1, \qquad C_- = 4n - 1 (u = 6n - 1), \qquad C_+ = 8n + 3 (u = 6n + 1),$$

and $C \equiv 3 \pmod 4$ always — for p, q coprime to 6, $q^2 - p^2 \equiv 0 \pmod{24}$, so $C = (q^2 - p^2)/6 - 1 \equiv -1 \pmod 4$ — an exact property that still gives no protective invariant, since a single line's deletion count has no fixed parity. The main term is $M = CP_2$.

A.2 Local deviations

The deviation $\varepsilon_r(u)$, the phase of a ruler inside a sector and its quadratic motion came here from the parent document. They belong to [P4, §3.4], where Theorem 3 uses them, and have been moved there. Nothing in this paper needs them.

A.3 The ladder in the width of the window, and the square-endpoint sharpening

The four results below sharpen Corollary 1 without strengthening it; their census is the largest computation in this paper. An earlier version had them in §2.2; they sit here so that §2.2 reads as Theorem 1, the criterion, and the dichotomy.

The six are the first rung of a ladder in the width of the window. The argument does not need the window to stop at $(M + 6)^2$. Widen it to $W_{M,k} = (M^2, (M + 6k)^2)$, keeping the lines at $p \le M$.

> **Corollary 1a (the *k*-ladder).** Let $M \equiv 3 \pmod 6$ and $k \ge 1$ satisfy $M > (6k)^2$, and switch on every line $p \le M$. Then every composite endpoint of an open cell of $W_{M,k}$ is a product $(M+a)(M+b)$ of two primes with $a \le b$ even, not divisible by 3, and $a + b \le 12k$; the number of such positions is $K_k = 4k + (4k - 2) + \cdots + 2 = 2k(2k + 1)$. Moreover **no open cell of $W_{M,k}$ has both endpoints composite**, so every exception is a P_2 with both factors above M sitting beside a genuine prime, and $T_{M,k} = C_{M,k} - X_{M,k}, \qquad X_{M,k} \le 2k(2k + 1)$.

Proof. A survivor has every prime factor above M, so a composite one has exactly two: three would exceed $M^3 > (M + 6k)^2$. Coprimality to 6 forces both offsets even and prime to 3.

For the range of the offsets, expand $n = M^2 + M(a+b) + ab$. If $a+b \le 12k$ then $ab \le \left(\frac{a+b}{2}\right)^2 \le (6k)^2$, with equality only at $a = b = 6k$, which is divisible by 3 and so excluded; hence $ab < 36k^2$ and $n < (M + 6k)^2$. Conversely $a+b$ is even, so if $a + b > 12k$ then $a + b \ge 12k + 2$ and

$$n - (M + 6k)^2 = M(a + b - 12k) + ab - 36k^2 \ge 2M - 36k^2 > M > 0$$

by the hypothesis $M > 36k^2$. So being in the window is *equivalent* to $a + b \le 12k$. Counting: a runs over the even non-multiples of 3 in $[1, 6k - 2]$, and for each the admissible $b \ge a$ with $b \le 12k - a$ number $4k, 4k - 2, \ldots, 2$, giving $K_k = 2k(2k + 1)$.

For the last claim, let $n = (M+a)(M+b)$ and $n' = (M + c)(M + d)$ be two such products. Then

$$n' - n = M\big((c + d) - (a + b)\big) + (cd - ab).$$

If $c+d = a+b$ the difference is cd-ab, divisible by 4 since all four offsets are even, so never 2. If $c+d \neq a+b$ the first term has absolute value at least $2M$ while $|cd-ab| < 36k^2 < M$, so $|n'-n| > M > 2$. So two composite survivors never share a cell. ■

Verification. At k=1 the six pairs are $(2,2),(2,4),(2,8),(2,10),(4,4),(4,8)$ — exactly E_M. The counts $K_k = 6, 20, 42, 72, 110, 156, 210, 272$ were checked against the definition for $k \leq 8$. Over every $M \equiv 3 \pmod 6$ below 1400 with $M > 36k^2$ and $k \leq 5$ — 689 windows — every exception matched the description and **no open cell had both endpoints composite**. The largest $X_{M,k}$ seen was 2 at k=1 against the ceiling 6, and 6 at k=3 against 42.

The hypothesis is load-bearing. Dropping $M > 36k^2$ on the same range gives both failures at once: open cells with two composite endpoints (1 at k=2, 7 at k=3, 15 at k=5) and exceptions outside the description (3, 13 and 30). The condition is stronger than needed — Theorem 1 needs none at k=1 — but it makes the short argument above work.

And widening does not help. The ceiling grows like $4k^2$ and the open cells only like $2kM/\log^2 M$, so the ratio $C_{M,k}/K_k$ scales as $M/((2k+1)\log^2 M)$: linearly worse in k. **The case $k = 1$ already in Theorem 1 is the best one**; the ladder completes the description and is no step toward forcing a survivor. The missing ingredient is unchanged: a lower bound on $C_{M,k}$.

Which old lines can close two of the six at once. Label the six cells by their composite member: $A = (M+2)^2$, $B = (M+2)(M+4)$, $C = (M+4)^2$, $D = (M+2)(M+8)$, $E = (M+2)(M+10)$ and $F = (M+4)(M+8)$. An old line r closes a cell when it divides either member. Then:

> **Corollary 1b.** Suppose $M+2$ and $M+4$ are both prime, so that all six cells exist. For $r > 31$, a line closes at most one of the six, with the single exception that $r \mid M+8$ closes D and F together.

Proof. Write $p = M+2$, so $M+4 = p+2$, $M+8 = p+6$ and $M+10 = p+8$. Since $M \equiv 3 \pmod 6$, each composite's residue fixes its partner: the six cells are (A, A-2), (B, B+2), (C, C-2), (D, D-2), (E, E+2), (F, F+2), and in terms of p the six partners are

$$Q_A = p^2 - 2, \quad Q_B = p^2 + 2p + 2, \quad Q_C = p^2 + 4p + 2,$$

$$Q_D = p^2 + 6p - 2, \quad Q_E = p^2 + 8p + 2, \quad Q_F = p^2 + 8p + 14.$$

Closures through the composite member. A line $r \leq M$ is smaller than both p and $p+2$, so r divides none of $A = p^2$, $B = p(p+2)$, $C = (p+2)^2$ except through $M+8$ or $M+10$. Hence $r \mid M+8$ closes D and F, and $r \mid M+10$ closes E; and r cannot divide both $M+8$ and $M+10$, whose difference is 2.

Two partners. If r divides Q_X and Q_Y it divides their difference. Nine of the fifteen differences are $2p$, $4p$, $6p$, $2q$, $6q$, $8q$ or 12 with $q = p+2$, and each forces $r \in \{2,3\}$ or $r \in \{p,q\}$, all excluded. The other six are settled by putting the resulting congruence back into one partner:

pair	r divides	substitution	conclusion
A,C	p+1	$Q_A \equiv 1 - 2$	$r \mid 1$
A,E	$2p$+1	$4Q_A \equiv 1 - 8$	$r \mid 7$
B,D	p-1	$Q_B \equiv 1 + 2 + 2$	$r \mid 5$
C,D	p-2	$Q_C \equiv 4 + 8 + 2$	$r \mid 14$
C,F	p+3	$Q_C \equiv 9 - 12 + 2$	$r \mid 1$
D,F	p+8	$Q_D \equiv 64 - 48 - 2$	$r \mid 14$

One composite and one partner. If $r \mid M + 8$ then $p \equiv -6$, and the six partners reduce to $34, 26, 14, -2, -10, 2$ modulo r, so r divides no partner beyond D and F themselves. If $r \mid M + 10$ then $p \equiv -8$ and the partners reduce to $62, 50, 34, 14, 2, 14$ — the fifth is $Q_E \equiv 64 - 64 + 2$, so E is closed through its composite member, not its partner. The largest new modulus is $62 = 2 \cdot 31$, the source of the constant. ■

The constant is sharp. The bound $r > 31$ cannot be lowered: $r = 31$ closes A and E together whenever $31 \mid M + 10$ (which already gives $31 \mid (M + 2)^2 - 2$, since $(-8)^2 - 2 = 62$). This happens in 128 of the twin sectors below $M = 400{,}000$, the first at $M = 1695$.

Verification. Zero violations over all 3,802 sectors below $M = 400{,}000$ with M+2 and M+4 both prime, so that all six cells exist. There the only multiple closure by a line above 31 is $\{D, F\}$, 4,060 times; below 31 every combination appears, the commonest being $\{D, E, F\}$, $\{B, D\}$, $\{B, E\}$ and $\{D, F\}$.

So the large lines cannot gather the exceptions: **collective covering of the six is the small lines' work.** This does not block the covering, and we say why so it is not attempted: at $M = 195$, where 197 and 199 are both prime and all six cells exist, three lines suffice — 151 closes A, 5 closes B and E, and 7 closes C, D and F. Lines above 31 alone can still close the six, using five of them.

A symmetric window in which one survivor suffices. The sector can be replaced by the window between the squares of a twin-shaped pair. Put $P = 6r$-1 and $Q = 6r$+1 and take

$$J_r = (P^2,\ Q^2),$$

which holds exactly $N = 4r$-1 complete cells ($6b$-1, $6b$+1) and is centred on $PQ + 1 = (6r)^2$ — verified for every $r < 3000$. Switch on every prime line $\ell < P$. A composite **inside** (P^2, Q^2) has a prime factor below P, except PQ itself. The open interval matters: P^2, on the lower boundary, is a composite with no prime factor below P, and it is not a cell member. So:

> **Corollary 1c.** In J_r, a surviving cell other than the central one is a twin pair; and if the **central** cell (PQ, PQ+2) survives, then P and Q are both prime, so (P,Q) is a twin pair. **One survivor of any kind suffices.**

Verification. Over $r < 400$: 12,242 survivors, each a twin by the relevant clause, no exceptions. In the first 1000 windows every window has a survivor, and the minimum survivor count is 2.

This is sharper than the sector form, where Corollary 1 lowers the sufficient count to $C_M \geq 3$; here it is 1. **The gain is in the statement, not in the difficulty:** one must still show that a set of lines fails to cover a set of cells, and the covering control of [P9, §3.5] applies unchanged.

Two facts about the window that narrow how a covering could be completed. A line $p = 6k \mp 1$ can close two cells only at separations $\{2k,\ 4k - 1\}$ or $\{2k,\ 4k + 1\}$, so no line closes two adjacent cells.

And a line $\ell > N$ that closes two cells not already closed does so with **prime** cofactors differing by 2 or 4: the window has length $24r$ and $\ell > 4r - 1$, forcing the cofactor gap below 6, and both cofactors are coprime to 6. Over the first 1000 windows: 303 such lines, 224 at gap 2 and 79 at gap 4, none otherwise and no composite cofactor.

And why this does not become a route. At $r = 1000$ the small lines leave 140 cells open; the 233 larger lines could close 466 between them, and close 18 — all singletons, leaving 122 survivors. The open cells grow like $r/\log^2 r$ and what the large lines close like $r/\log^3 r$, while the crude bound on their capacity is $4r/\log r$: **the separation is real and lies between two logarithms, where no counting bound reaches it.** Moving the split point from $X^{0.235}$ to $X^{0.49}$ does not bring the crude bound below the count of open cells at any depth.

Removing the two square positions, and what is left. Two of the six carry perfect squares, $A = (M+2)^2$ and $C = (M+4)^2$. These are the only cells of the sector with a square endpoint: the squares between M^2 and $(M+6)^2$ are $(M+j)^2$ for $1 \le j \le 5$, and only $j = 2$, 4 give a member coprime to 6. A twin has two prime members, so **discarding the cells with a square endpoint discards no twin.** Write C_M° for the number of open cells with neither member a perfect square.

> **Corollary 1d.** Suppose $(M{+}2,\ M{+}4)$ is not a twin pair. Then at most **one** of the four non-square positions B, D, E, F is open, so $C_M^\circ \ge 2 \implies$ the sector contains a twin. Moreover the line 5 alone settles two residue classes of M: if $M \equiv 21 \pmod{30}$ then none of B, D, E, F is open, whatever $M{+}2$ and $M{+}4$ are, and if $M \equiv 27 \pmod{30}$ then only B can be, so in both classes $C_M^\circ \ge 1 \implies$ the sector contains a twin, or $(M+2, M+4)$ is one.

Proof. By the grouping in Corollary 1, B needs both $M{+}2$ and $M{+}4$ prime and so is closed under the hypothesis; D, E need $M{+}2$ prime and F needs $M{+}4$ prime, so F cannot be open with D or E. It remains to rule out D and E together. Write $p = M{+}2$. The cell of D is $\big(p(p+6) - 2,\ p(p+6)\big)$, with members $p(p{+}1)$ and $(p{+}2)(p{\text{-}}1)$ modulo 5; both are non-zero only for $p \equiv 2 \pmod 5$. But then $5 \mid p + 8$, so the line 5 closes the composite member $p(p{+}8)$ of E.

For the two classes, the same computation for all four positions gives survival modulo 5: B is open only at $p \equiv 4$, D only at $p \equiv 2$, E only at $p \equiv 1$, and F only at $p \equiv 0, 1, 2$. So $p \equiv 3$ closes all four and $p \equiv 4$ leaves only B; with $p = M{+}2$ and $M \equiv 3 \pmod 6$ these are $M \equiv 21$ and $M \equiv 27 \pmod{30}$. ■

Verification. Over the same 3,332 sectors: D and E are never open together; in the 2,992 sectors with a non-twin root pair, open non-square positions never exceed 1; in all 666 sectors with $M \equiv 21 \pmod{30}$, root twin or not, there are 0; and in the 666 with $M \equiv 27 \pmod{30}$ never more than 1.

The same census to $M = 10^9$. Corollary 1a makes the test cheap. No two of the six candidate composites differ by 2, so the partner of an exception is never itself an exception; hence for $M > 36$ a partner is rough exactly when it is prime. A composite $(M{+}a)(M{+}b)$ is rough exactly when $M{+}a$ and $M{+}b$ are both prime. So primality alone decides openness:

$$A:\ M{+}2,\ (M{+}2)^2{-}2; \qquad B:\ M{+}2,\ M{+}4,\ (M{+}2)(M{+}4){+}2;$$

$$C:\ M{+}4,\ (M{+}4)^2{-}2; \qquad D:\ M{+}2,\ M{+}8,\ (M{+}2)(M{+}8){-}2;$$

$$E:\ M{+}2,\ M{+}10,\ (M{+}2)(M{+}10){+}2; \qquad F:\ M{+}4,\ M{+}8,\ (M{+}4)(M{+}8){+}2.$$

Over all 166,666,665 sectors below 10^9: ***D* and *E* are never open together; a sector whose root pair is not a twin never has more than one open non-square position; there are**

none in any sector with $M \equiv 21 \pmod{30}$; and there is $\{B\}$ or nothing in every sector with $M \equiv 27 \pmod{30}$. The counts by decade, with the two sifting dimensions read off the densities:

decade of M	sectors	A	C	B	D	E	F	sq. $\times \log^2$	non-sq. $\times \log^3$	ratio $\times \log$
10^4–10^5	1.5×10^4	655	681	70	46	58	148	9.498	23.658	2.491
10^5–10^6	1.5×10^5	4,466	4,487	383	289	291	738	9.525	22.843	2.398
10^6–10^7	1.5×10^6	32,150	32,216	2,326	1,741	1,767	4,931	9.573	23.940	2.501
10^7–10^8	1.5×10^7	244,729	243,890	15,391	11,176	11,708	32,167	9.686	24.095	2.488
10^8–10^9	1.5×10^8	1,912,716	1,910,857	106,290	76,561	81,072	222,588	9.742	24.249	2.489

The last three columns are the point. They give the square family's density against $\log^2 M$, the non-square family's density against $\log^3 M$, and their ratio against $\log M$. The square positions have dimension 2 and the non-square ones dimension 3, so all three should tend to constants. All three do: the first settles near 9.7 after a slow climb, the second near 24.2, and the third is 2.49 across four consecutive decades, varying by under half a per cent. **The dimension count of Corollary 1e is thus measured, not only derived.** In the last decade the non-square exceptions occupy 0.324% of sectors, so a single open non-square cell forces a twin in 99.68% of them.

One asymmetry inside the family: F is about twice as common as B — the ratio is 2.09 in each of the last two decades and 2.12 in the one before them — though the two carry linear conditions of the same strength, since prime pairs at gap 4 and at gap 2 have the same singular series. The difference lies wholly in the partner: the prime factors of $Q_B = (M+3)^2 + 1$ must be 1 mod 4, those of $Q_F = (M+6)^2 - 2$ must be ± 1 mod 8, and the two densities differ by that factor.

How rare the non-square exceptions are, and in what sense this can be proved. The four non-square positions are sifted in dimension **three**, not two. The composite member of each is a product of two linear forms in M, so it forbids two residues modulo every line r; the partner is an irreducible quadratic and forbids two more or none. With $\left(\frac{\cdot}{r}\right)$ the Legendre symbol, the partners are

$$Q_B = (M+3)^2 + 1, \quad Q_D = (M+5)^2 - 11, \quad Q_E = (M+6)^2 - 14, \quad Q_F = (M+6)^2 - 2,$$

so the number of residues of M that the line r forbids is

$$3 + \left(\tfrac{-1}{r}\right), \quad 3 + \left(\tfrac{11}{r}\right), \quad 3 + \left(\tfrac{14}{r}\right), \quad 3 + \left(\tfrac{2}{r}\right)$$

respectively, each of mean 3 since the characters are non-principal. This extends Appendix B.1, which gives dimension 2 for A, C and dimension 3 for D, E, F, to the last position B.

Let $U(y)$ be the density of residue classes of M in which at least one of B, D, E, F survives every line $r \le y$. Then $U(y) \asymp (\log y)^{-3}$; exact inclusion–exclusion over the four positions gives

lines up to y	23	97	199	499	997	4,999	19,997
classes free of all four	79.57%	91.43%	94.41%	96.40%	97.32%	98.55%	99.07%
$U(y)(\log y)^2$	2.01	1.79	1.57	1.39	1.28	1.05	0.91
$U(y)(\log y)^3$	6.30	8.20	8.29	8.64	8.84	8.96	9.02

The third row settles the order: the second row falls throughout, the third steadies near 9.

> **Corollary 1e.** The number of $M \le X$ with $M \equiv 3 \pmod 6$ whose sector carries an open non-square exception is $o(X)$.

Proof. Fix y. Every line $r \le y$ is active in a sector with $M > y$, so an open non-square exception there survives all of them, and its residue class of M lies in the exceptional set counted by $U(y)$. That set is a union of classes modulo a fixed modulus, so the density of such M up to X tends to at most $U(y)$ as $X \to \infty$. Then letting $y \to \infty$, with $U(y) \to 0$, gives the claim. ■

The order of the limits matters: the modulus of the class decomposition grows with y, so one may not put $y = M$ and read off a rate. The statement as proved gives vanishing density and no rate.

A rate is available, from a standard sieve theorem rather than this argument. For $M > 36$, an exception is open exactly when the polynomials in its row of the classification above are all prime: A needs $M+2$ and $(M+2)^2-2$; C needs $M+4$ and $(M+4)^2-2$; B needs $M+2$, $M+4$ and $(M+2)(M+4)+2$; D needs $M+2$, $M+8$, $(M+2)(M+8)-2$; E needs $M+2$, $M+10$, $(M+2)(M+10)+2$; F needs $M+4$, $M+8$, $(M+4)(M+8)+2$. The linear members must be prime, since a composite one would have a prime factor $\le M$. The partner must be prime, since otherwise the cell would carry two surviving composites, which the six-position classification excludes. With $M = 6t+3$, each row is a set of distinct irreducible polynomials in t with positive leading coefficients — the quadratic discriminants are $288, -144, 288, 1584, 2016, 288$, none a square — with two members in rows A, C and three in the others. The polynomial Selberg upper bound [13, eq. (2)] then gives $O(Y/(\log Y)^g)$ values of $t \le Y$ at which all g are prime. A row with a fixed prime divisor would instead have a finite solution set, so no admissibility hypothesis is needed. Summing over the rows:

> **Corollary 1e' (quantitative form).** $\operatorname{card}\{M \le X : \text{the sector carries an open non-square exception}\} \ll \dfrac{X}{(\log X)^3}$, and with the square positions included the bound is $O(X/(\log X)^2)$. No hypothesis on the root pair is required.

Equivalently, since $C_M - T_M$ is the number of open exceptions and is at most six,

$$0 \le \sum_{\substack{9 \le M \le X \\ M \equiv 3 \ (6)}} (C_M - T_M) \ll \frac{X}{(\log X)^2}, \qquad 0 \le \sum (C^\circ_M - T_M) \ll \frac{X}{(\log X)^3},$$

C°_M counting open cells with no square endpoint. **So outside $O(X/\log^3 X)$ roots up to X, the square-discarded criterion has no exceptions at all: $C^\circ_M = T_M$.** This applies a published sieve bound; it is not a new sieve, and it gives no survivor in any prescribed sector — both sides may be zero. Nor do separate upper bounds for the two families show that almost every actual exception is a square one; that would need control of the denominator.

A statement about the partners that the literature does supply. With $M = 6t+3$, each of the six partners is a quadratic in t:

type	partner as a quadratic in t	discriminant
A	$36t^2 + 60t + 23$	288
B	$36t^2 + 72t + 37$	-144
C	$36t^2 + 84t + 47$	288
D	$36t^2 + 96t + 53$	1584
E	$36t^2 + 108t + 67$	2016
F	$36t^2 + 108t + 79$	288

No discriminant is a perfect square, so all six are irreducible over $\mathbf{Z}$; the leading coefficient is positive; and none has a fixed prime divisor. Modulo 2 and 3, the 36 vanishes and leaves the constant term, odd and prime to 3 in every row; for $p > 3$ the reduction stays a true quadratic, with at most two roots. (Checked directly for every prime up to 73; the greatest common divisor of the first sixty values is 1 in each row.) Iwaniec [7] proves that an irreducible quadratic with positive leading coefficient and no fixed prime divisor takes values with at most two prime factors infinitely often, and Lemke Oliver [8] gives the form with the values also square-free. Hence:

> **Remark.** For each of the six types separately, there are infinitely many t at which that partner is square-free with at most two prime factors.

Three limits keep this a remark. It is about **each partner separately**: six infinite sets of t, which need not meet, while an open cell needs its conditions at one and the same t. It does not separate "prime" from "product of two primes", the distinction an exception turns on. And it says nothing about the composite member, whose openness needs $M+a$ and $M+b$ both prime. It is a correct application of a published theorem to the objects of §2.2, and no more.

This is a change of bookkeeping, not of strength. At most two square cells are discarded, so $C^{\circ}_M \geq C_M - 2$ and the criterion still amounts to $T_M \geq C_M - 2$, which is Corollary 1. What it adds: the loss is now located exactly — two square positions and one other — and the proof needs no scan of residue classes. The gain in the two classes of the second clause is real, and covers two fifths of all sectors.

And what it does not buy. C_M exceeds the twin count of the sector by at most six. Any lower bound on C_M is thus a lower bound on twins, and (2.3) is an exact reformulation, not a route. We record it as the sharpest form of the twin criterion the framework has produced, not because it weakens the problem.

Appendix B — Routes that were tried and closed

An earlier version had these four accounts in §2. They sit here so that §2 reads as a sequence of results. They are kept in full because a route closed by measurement or proof is worth more written down than tried again. The stub in §2 states each conclusion and its status as proof or measurement.

B.1 Blocks of consecutive periods, and why lengthening the block does not help

Section 2.3 bounds the exceptions over one period of 35 sectors. Consecutive periods are coupled just as consecutive sectors are, so one can ask the same of a block of L periods — $35L$ sectors — and the answer moves. Write B_L for the maximum of $\sum_i |S_{M_i} \cap E_{M_i}|$ over such a block under the twinless hypothesis, so $B_1 = 31$ is §2.3 and

$$\sum_i C_{M_i} > B_L \implies \text{a twin in the block.}$$

How B_L is computed, and a trap worth naming. The natural procedure — list the maximal configurations of each residue class and test each for a fixed prime divisor — is *not* sound here. A class whose maximum is m also carries configurations of size m-1, m-2 and so on, got by deleting cells; deleting a cell removes conditions and can only ease admissibility. Those sub-configurations are never listed, so "every configuration at level ℓ dies" proves nothing about level ℓ. The sound procedure reverses the order: fix a residue for each prime q, switch the line q on *inside* the transfer recursion, and read off the maximum. By construction that is the largest admissible configuration at those residues, and maximising over residues gives B_L. It is also faster by orders of magnitude.

Values. By that procedure, maximising over the residues of every prime up to the stated bound:

$$B_1 = 31, \qquad B_3 = 67, \qquad B_5 = 100, \qquad B_7 = 138, \qquad B_9 = 163.$$

The first three are certain: B_1 is §2.3; B_3 and B_5 are stable under every prime tested, and B_5 was obtained twice, by the sound procedure and by exhaustive enumeration with an independent admissibility test, which agree. B_7 and B_9 are stable across twelve and eight consecutive primes but were computed with pruning, so they are lower bounds with strong stability, not certified maxima.

The extremal configuration at L=3 carries 66 linear and 67 quadratic conditions — 133 polynomials of total degree 200 — and every prime $q \le 200$ leaves it an admissible residue. So §2.3's conclusion transfers: no finite set of lines lowers 67 either. Two features of the optimum stand out. First, it does *not* keep the single-window maximum in the middle: the three windows contribute 20, 26, 21, so the block optimum gives up 31 - 26 = 5 of the central window's capacity to gain elsewhere. Second, the killing is concentrated: at every level from 144 down to 138 at $L = 7$, the primes 29 and 37 remove about ninety per cent of the configurations, the rest falling to 43, 53, 71, 73, 79 and 97.

In requirement per sector, $(B_L + 1)/35L$ reads 0.914, 0.648, 0.577, 0.567, 0.521. It falls; the next paragraph says why, and how far.

The quadratic shadow, and the order of B_L. Besides its linear primality conditions, each exceptional cell needs the *other* member of the cell to survive. Those partners are

$$Q_A = (M{+}2)^2{-}2, \quad Q_C = (M{+}4)^2{-}2, \quad Q_D = (M{+}5)^2{-}11, \quad Q_E = (M{+}6)^2{-}14, \quad Q_F = (M{+}6)^2{-}2,$$

so modulo a prime $q > 17$ each type forbids exactly

$$\omega_A = \omega_C = 2 + \chi_2(q), \qquad \omega_D = 3 + \chi_{11}(q), \qquad \omega_E = 3 + \chi_{14}(q), \qquad \omega_F = 3 + \chi_2(q),$$

residue classes, with $\chi_d(q)$ the Legendre symbol. The linear conditions give one forbidden class for A and C and two for D, E and F; the partner gives two more exactly when the relevant d is a quadratic residue. The quadratic characters are non-principal, so each ω has mean 2 for A and C and 3 for D, E and F.

Proposition 2. Let a block of $N = 35L$ consecutive sectors carry an admissible configuration of exceptional cells under the twinless hypothesis, and for $X \in \{A, C, D, E, F\}$ let I_X be the set of sector indices carrying a cell of type X. Then $|I_A|, |I_C| \ll \dfrac{N}{\log^2 N}$, $\quad |I_D|, |I_E|, |I_F| \ll \dfrac{N}{\log^3 N}$, and consequently $B_L \ll \dfrac{L}{\log^2 L}$.

Proof. **Step 1: each type forbids an exact number of residue classes.** Fix a prime $q > 17$. Admissibility at q means M_0 can take some residue modulo q at which no polynomial of the configuration vanishes; fix it and write $t \equiv M_i \pmod q$. As 6 is invertible modulo q, the map $i \mapsto t$ is a bijection of $\mathbb{Z}/q$, so it suffices to count forbidden values of t.

A cell of type A at index i carries one linear condition and one quadratic one:

$$M_i + 2 \text{ prime}, \qquad (M_i + 2)^2 - 2 \text{ surviving}.$$

Modulo q these forbid

$$t \equiv -2 \qquad (\text{one class}), \qquad (t+2)^2 \equiv 2 \qquad (1 + \chi_2(q) \text{ classes}),$$

with $\chi_d(q)$ the Legendre symbol. The two never coincide, since $t \equiv -2$ would give $0 \equiv 2$. So exactly $2 + \chi_2(q)$ classes, and the same for C with -2 replaced by -4.

A cell of type D carries two linear conditions and one quadratic:

$$M_i + 2 \text{ prime}, \qquad M_i + 8 = M_{i+1} + 2 \text{ prime}, \qquad (M_i + 5)^2 - 11 \text{ surviving}.$$

The linear classes $t \equiv -2$ and $t \equiv -8$ are distinct for $q > 3$, and

$$(t+5)^2 \equiv 11$$

adds $1 + \chi_{11}(q)$ classes, disjoint from them since either substitution gives $9 \equiv 11$, impossible for $q > 2$. So exactly $3 + \chi_{11}(q)$.

The types E and F go the same way. Collecting the four computations, with $\omega_X(q)$ the number of forbidden classes:

type	linear offsets	quadratic condition	$\omega_X(q)$
A	-2	$(t+2)^2 \equiv 2$	$2+\chi_2(q)$
C	-4	$(t+4)^2 \equiv 2$	$2+\chi_2(q)$
D	-2, -8	$(t+5)^2 \equiv 11$	$3+\chi_{11}(q)$
E	-2, -10	$(t+6)^2 \equiv 14$	$3+\chi_{14}(q)$
F	-4, -8	$(t+6)^2 \equiv 2$	$3+\chi_2(q)$

In every case the overlap check between linear and quadratic classes reduces to $16 \equiv 14$ or $4 \equiv 2$, impossible for $q > 2$, so the counts simply add.

Verification. Computed directly for all 423 primes $19 \le q < 3000$: zero disagreement with $\omega_A = \omega_C = 2+\chi_2$, $\omega_D = 3+\chi_{11}$, $\omega_E = 3+\chi_{14}$, $\omega_F = 3+\chi_2$.

Step 2: the sifting dimension is the mean of ω_X. I_X lies in $\{0, 1, \dots, N-1\}$ and avoids $\omega_X(q)$ classes modulo every prime $q > 17$; the primes $q \le 17$ give a bounded factor and are ignored. As χ_2, χ_{11} and χ_{14} are real non-principal characters — of conductors 8, 44 and 56 — the series $\sum_q \chi_d(q)/q$ converges, as $L(1,\chi_d)$ does not vanish. Hence

$$\prod_{17<q<z}\Big(1-\frac{\omega_X(q)}{q}\Big) = \prod_{17<q<z}\Big(1-\frac{\kappa_X}{q}\Big)\cdot\prod_{17<q<z}\frac{1-\omega_X(q)/q}{1-\kappa_X/q} \asymp \frac{1}{(\log z)^{\kappa_X}},$$

with $\kappa_A = \kappa_C = 2$ and $\kappa_D = \kappa_E = \kappa_F = 3$, the second product converging because its logarithm is $-\sum_q \chi_d(q)/q + O(\sum_q q^{-2})$.

Step 3: the upper-bound sieve. Selberg's sieve applied to the interval $\{0, \dots, N-1\}$ with the removed classes of Step 1 gives, for any z,

$$|I_X| \ll N\prod_{q<z}\Big(1-\frac{\omega_X(q)}{q}\Big) + O(z^{2+\varepsilon}).$$

(The implied constant is the sieve constant of the quoted theorem. A leading factor $1+o(1)$ on the raw Euler product is not what a fixed-level Selberg bound gives; an earlier version printed it that way. Only the order of the main term is used below.) The error is the usual one for sifting an interval: with $|\lambda_d| \le 1$ the remainder is $\sum_{d_1,d_2<z}|r_{[d_1,d_2]}|$, where r_m counts the removed classes modulo m, so it is at most a divisor-function factor times z^2.

Taking $z = N^{1/3}$ makes the error $O(N^{(2+\varepsilon)/3})$ — not $O(N^{2/3})$, as an earlier version had it — which for any fixed $0 < \varepsilon < 1$ is still $o\big(N/(\log N)^{\kappa_X}\big)$. By Step 2 the main term is $\ll N/(\log N)^{\kappa_X}$. This is the stated bound for each I_X.

Step 4: summation. Under the twinless hypothesis type B is empty, since it needs both $M+2$ and $M+4$ prime. A sector carries at most one cell of each type, so the block holds exactly $\sum_X |I_X|$ exceptional cells, which by Step 3 is $\ll N/(\log N)^2$. With $N = 35L$ this is $\ll L/\log^2 L$. ■

This is a computation, not a new tool, and should be read as one. Reading the sieve dimension off the number of roots of the defining polynomials modulo each prime is the standard definition of dimension — the condition $\sum_{q<s}(\log q)/f(q) = \kappa \log s + O(1)$ of Halberstam and Richert [6]. Passing from a dimension-κ sifting density to an upper bound of order $N/(\log N)^\kappa$ is

the standard Selberg estimate; see [2] for the higher-dimensional theory and [6] for the sieve itself. The twin problem in this language is the case $\rho(q) = 2$, dimension 2. Only the input is particular to this setting: the exceptional cells of Theorem 1 have the five partners above, so the quadratic characters χ_2, χ_{11} and χ_{14} appear and split the five types into dimensions 2,2,3,3,3. The conclusion then follows from the standard machinery, for which we claim nothing.

Two remarks on its shape. The bound is an **upper-bound sieve only**, and upper bounds in a fixed dimension are the direction where sieve methods have no parity obstruction; nothing here bears on the lower-bound side. And the two-sector types are smaller than the one-sector types by a full logarithm, so in a long block almost every exception is of type A or C — a longer block is structurally *simpler*, not more complicated.

The first consequence is the one that matters here: **the density of exceptions tends to zero**, so the requirement per sector has no positive lower bound.

And the bound is numerically empty over every computable range. Taking $\rho^*(D) \sim D/\log D$ for the largest admissible set in an interval of diameter D, the crude form $B_L \le 2\rho^*(210L)-1$ gives, per sector, 2.22 at L=1, 1.59 at L=9, 1.06 at L=401, 0.82 at $L = 10^4$ and 0.51 at $L = 10^8$. It falls below 1 only near $L \approx 400$ and reaches 1/2 only near $L \approx 10^8$. The measured values are far below it throughout, so the observed decrease is not this bound at work. Nor do the data fix the exponent: over $1 \le L \le 101$ the quantity $B_L \log^2(210L)/(210L)$ reads $4.22, 4.42, 4.61, 4.99, 4.91, 5.80, 6.68, 7.37$ — rising, not settling — while $B_L \log(210L)/(210L)$ reads $0.789, 0.685, 0.663, 0.685, 0.651, 0.691, 0.720, 0.740$. On this range $\log D$ moves only from 5.3 to 10.0, too little to separate the two laws.

What the whole calculation does not buy, stated exactly. Since $B_L = o(L)$, the criterion no longer asks for a survivor in every sector or every second one: it suffices to prove $\sum_i C_{M_i} \ge \varepsilon L$ for any fixed $\varepsilon > 0$, that is, that a positive proportion of sectors holds at least one open cell. At $M_0 = 448{,}353$ the measured survivor total is 366,120, 1,100,106 and 3,304,035 over $L = 1$, 3, 9 — 10,461, 10,477 and 10,489 per sector, flat — so the inequality holds by a factor of 11,810, 16,419 and 20,270, widening with L. The survivor total is linear in L with a large constant; its logarithm is $\log M_0$, not $\log L$, so no lengthening changes its shape.

> **The reduction is therefore real and it is not a route.** The survivor side now needs not a growth rate but an *existence* statement: one open cell in a positive proportion of sectors. By Theorem 1 an open cell is a twin pair unless it sits at one of six named places, and [P9, §2.3] measures that discrepancy at 0, 1 and 3 over three full periods. So "prove $\sum_i C_{M_i} \ge \varepsilon L$" is "prove that a positive proportion of sectors contains a twin". Shrinking the right-hand side from 32 to $O(L/\log^2 L)$ changes the number on the right and leaves the missing ingredient on the left exactly as it was.

B.2 The square-phase mean, and the end of the "poor window" question

Are windows anchored at squares systematically poorer than windows placed anywhere? The answer is an identity, not an experiment.

For a window starting at r^2, the pair at offset j survives q exactly when $r^2 \not\equiv -2j$ and $r^2 \not\equiv -2j-2 \pmod q$. With $\rho_q(a) = \mathrm{card}\{r : r^2 \equiv a\}$ and $\nu_q(j) = \rho_q(-2j) + \rho_q(-2j-2)$ — the two conditions

cannot hold at once — the average μ_{sq} over **all square phases** is exactly

$$\mu_{\mathrm{sq}}(L) = \sum_{j<L} \prod_{q} \frac{q - \nu_q(j)}{q}. \qquad \text{(B.2.1)}$$

Against the naive density prediction $E = L\prod(q-2)/q$, and the restricted average μ_{cop} over roots coprime to every old line:

p	T_p^- (actual)	μ_{sq}	μ_{cop}	$E = L\delta$
11	2	2.943	2.833	3.286
29	2	4.149	4.201	4.206
53	2	5.434	5.283	5.457
101	7	7.653	7.473	7.774
499	13	21.237	21.319	21.274
997	38	34.591	34.534	34.608

μ_{sq} agrees with E within about 1% from $p = 29$ on, and restricting the roots changes nothing.

> **The square-phase mean has the exact expression (B.2.1); numerically it tracks the generic density prediction closely, the two differing by about 1% or less from $p = 29$ onward in the sample above.** So the exact quantity has a closed form, and on the tested range it shows no bias in the mean. This replaces the earlier control experiment with a computation, though not with a proof that the two agree in the limit. Single windows are far from the mean — $T_{53}^- = 2$ against $\mu_{\mathrm{sq}} = 5.43$ — but the scatter is governed by the correlation function of [P3, Thm 4], not by any property of squares.

B.3 Two constructions that do not help

A window balanced across a bundle of lines. One can design the window instead of inheriting it. For every line of a bundle $p_{\min} \le p \le p_{\max}$ to make exactly $2r+1$ strikes about a common centre A divisible by all of them, the half-width H must satisfy $2rp_{\max} \le H < 2(r+1)p_{\min}$; so such a window exists iff

$$\frac{p_{\max}}{p_{\min}} < 1 + \frac{1}{r}.$$

For $r = 1$ that is $p_{\max} < 2p_{\min}$ — the lines 11,13,17,19 with $H = 40$ give three strikes each — and for $r = 2$ it is $p_{\max} < 1.5p_{\min}$. The side strikes of different lines never coincide while $r < p_{\min}$, so the bundle meets only at A.

Why it changes nothing. Take $p = 1009$, $q = 1013$, $r = 52$, so each makes 105 strikes in one window, and slide the window along the common multiples $A_t = pq(2t+1)$. The incidence census of those 105 positions against the lines 3, 5, 7 is $48, 24, 12, 8, 6, 4, 2, 1$ — the same in every window over $t = 0, \ldots, 11$, and the same as the census of the odd numbers modulo 210. More: the **word** itself, the sequence of incidence patterns across the 105 positions, is a cyclic rotation of the first window's word in each of 20 windows tested. The whole object is one word of length 105 and a rotation,

because 105 consecutive positions cover every residue modulo 105 once. The construction promised to separate the *count* of overlaps from their *arrangement*; the measurement says there is nothing to separate, since the arrangement is the same word shifted.

Ownership and the cofactor. Write a strike in a sector $[P^2, Q^2)$ as $N = \mathrm{pm}$. It is new for p exactly when $\mathrm{spf}(m) \geq p$, and the higher lines that return to it are exactly the primes q with $p < q \leq P$ and $q \mid m$. Two clean consequences: if $q > p$ shares a new point of p then $p^2 q < Q^2$, and conversely $P^2 \leq p^2 q < Q^2$ makes $p^2 q$ a guaranteed shared point. And if $p^3 \geq Q^2$ then $m < Q^2/p \leq p^2$ while m has no factor below p, so m is prime: in that layer every new strike is a product of exactly two primes (verified with no exception in the sectors $101 \to 103$, $499 \to 503$, $997 \to 1009$).

Why this too changes nothing. The condition $p^2 q_1 q_2 \cdots < Q^2$ is **not** a bound on interaction depth, and the derived layers $Q^{1/2}, Q^{2/5}, \ldots$ do not exist: $10387 = 13 \cdot 17 \cdot 47$ lies in $[101^2, 103^2)$ and $1009091 = 97 \cdot 101 \cdot 103$ in $[997^2, 1009^2)$. More to the point, even in the clean layer "is this strike new?" becomes "is the cofactor prime?", and for smaller p "is the cofactor free of prime factors below p?" — Buchstab's decomposition in the vocabulary of lines. The numbers agree: of the raw strikes of the clean layer, the new ones are 29 of 73, 1158 of 4756 and 1492 of 7938 in those three sectors — 40, 24 and 19 per cent and falling. Even where no higher line can return, most strikes are still repeats from below.

B.4 Smoothing the window with the classical kernels

Proposition 1 is stated for a sharp window, which pays a boundary error of order 1 per residue class. A Fejér kernel in place of the window removes most of that error, and moving the kernel removes more. **The smoothing itself is classical**, and so is the order it gains. Recorded below are an exact discrepancy formula for the single kernel, a bound for the moving one, and a mirror law for the sector index from the quadratic path of [P2, §4.5]. The point of the subsection, though, is the measurement that follows: how far this smoothing reaches inside the hierarchy of §2.9. It does not reach the part that decides the answer.

A mirror law for the sector index. Fix an odd modulus d and let $\rho(d)$ be the number of residues it forbids (2^i when d is a product of i lines). With $N_d(s)$ the number of those residues met inside the single sector $I_s = [a_s, a_{s+1})$, put $e_d(s) = N_d(s) - \rho(d) L_s/d$.

> **Proposition 3.** $e_d(s+d) = e_d(s)$; and for $0 \leq s \leq d-2$, $e_d(d-2-s) = -e_d(s)$, with $e_d(d-1) = 0$.

Proof. Periodicity is $a_{s+d} \equiv a_s \pmod{d}$, which is (4.4) of [P2, §4.5]. For the reflection, the two facts needed are

$$a_{d-2-s} \equiv a_{s+1} \pmod{d}, \qquad L_s + L_{d-2-s} = 12d.$$

The first says the mirror sector begins, modulo d, exactly where I_s ends. And $L_s + L_{d-2-s} = 12(s+1) + 12(d-1-s) = 12d$, so together the two sectors cover exactly twelve full periods of d and their errors cancel. The sector $s = d\text{-}1$ alone has length $12d$, so $e_d(d-1) = 0$. ■

Verification. Zero failures over the whole cycle for $d = 5, 7, 11, 13, 35, 55, 77$ and 385, with the two ingredients of the proof also checked separately. **So the word of phase errors along a cycle reads $e_0, e_1, \ldots, -e_1, -e_0, 0$: it is odd about its own centre.**

The triangular weight, and an exact discrepancy formula. Replace the indicator of a window by the triangular weight of half-width H,

$$w_H(u) = \max\Big(1 - \frac{|u|}{H},\ 0\Big),$$

the Cesàro weight, which is the Fejér kernel on $\mathbb{Z}/d\mathbb{Z}$: $H\ w_H$ is the self-convolution of the indicator of $\{0, \dots, H-1\}$, so its transform is $|\widehat{\mathbf{1}}|^2 \geq 0$ and decays like h^{-2}, not h^{-1}. The weight and that representation are standard in this very setting; see [1], where sieve functions are averaged over unions of residue classes in a short interval with the same weight.

Proposition 4. Write $H = \mathrm{q}d + s$ with $0 \leq s < d$. Then $\max_{b \bmod d} \Big| \sum_{n \equiv b \ (d)} w_H(n) - \frac{H}{d} \Big| = \frac{s(d-s)}{dH} \leq \frac{d}{4H}$. (B.4.1) with the maximum attained at $b = 0$.

Proof. Let c_j be the number of $n < H$ in the class $j \pmod d$ and $c_j = H/d + \delta_j$. The weighted count is $H/d + H^{-1}\sum_j \delta_j \delta_{j-b}$. Exactly s classes have $c_j = q+1$ and the rest $c_j = q$; those s classes form a cyclic interval A, so $\delta_j = \mathbf{1}_A(j) - s/d$ and

$$\sum_j \delta_j \delta_{j-b} = |A \cap (A+b)| - \frac{s^2}{d}.$$

This is largest at $b = 0$, where $|A \cap A| = s$ and the value is $s - s^2/d = s(d-s)/d$. Since (B.4.1) is a maximum of an **absolute** value, the negative extreme must be bounded too, and it is smaller. By inclusion–exclusion inside $\mathbb{Z}/d$, $|A \cap (A+b)| \geq \max(0,\ 2s-d)$, so the quantity is at least $-s^2/d$ when $s \leq d/2$ and at least $-(d-s)^2/d$ when $s \geq d/2$. In the first case $s^2 \leq s(d-s)$ and in the second $(d-s)^2 \leq s(d-s)$, so in both the $b = 0$ value dominates in modulus and the maximum in (B.4.1) is attained there. (Checked directly for every pair $d < 80$, $0 \leq s < d$.) Finally $s(d-s)/d \leq d/4$, with equality only when $s = d/2$, so for the odd moduli of this paper the second inequality in (B.4.1) is always strict. ■

Where this sits. Fejér smoothing and the resulting order $O(d/H)$ are classical: [9, Ex. 24.2.1.1(d)] asks for the pointwise estimate $0 \leq \Delta_N(x) \leq \min\{N, 1/(4N\|x\|^2)\}$, with the constant $\frac{1}{4}$ arising the same way, from $\sin \pi x \geq 2\|x\|$. But that estimate is not (B.4.1). The left side of (B.4.1) is an *average* of the kernel over a subgroup of frequencies, $d^{-1}\sum_{h \neq 0} e(-hb/d)\Delta_H(h/d)$, not a pointwise value. Applying the pointwise bound term by term and summing recovers the order but overshoots $d/(4H)$ by a factor of 2.93, 3.25 and 3.29 at $d = 11, 101, 1001$, stable in H. **The equality in (B.4.1) is the part we have not found stated anywhere.** It explains the near-sharpness of $d/(4H)$ without measurement: the ratio of the two sides is $4s(d-s)/d^2$, which for odd d is largest at $s = (d \pm 1)/2$ and equals $(d^2-1)/d^2$. The closest treatment of the same kernel on the same group we have found is Grieshop [5]. His result has a different shape — an $L^2 \to L^\infty$ bound of order $\sqrt{N/r}$ for an arbitrary subset — and his Remark 5.4 names the boundary error of size 1 that (B.4.1) evaluates, then sets it aside. We state (B.4.1) as elementary and probably folklore, not as new, and would be glad to be pointed to a source.

Moving the window as well: a double tent. Now slide the centre linearly, $c_t = c_0 + t$, and weight the times by a second triangle $q_T(t) = T^{-1}(1 - |t|/T)$.

Proposition 5. With that weighting, $|E_{d,b}| \leq \frac{d^4 + 10d^2 - 11}{45\ d\ H\ T^2} \ll \frac{d^3}{45\ H\ T^2}$, using the exact identity $\sum_{h=1}^{d-1} \csc^4(\pi h/d) = (d^4 + 10d^2 - 11)/45$.

Proof. Each triangle contributes the square of a Dirichlet kernel to the Fourier expansion of the error, giving $d^{-1}\sum_{h\neq 0}|D_H(h)|^2H^{-1}|D_T(h)|^2T^{-2}$; bounding $|D_N(h)|\le \csc(\pi h/d)$ and summing by the identity gives the statement. ■

The identity is classical, one of a family of finite cosecant power sums going back to Euler; see [3] for the $\csc^4$ case and its higher analogues. The bound holds with worst ratio 0.8958 over all (d,H,T,b) we tested. **Taking $H \asymp T \asymp r$ gives $|E| \ll (d/r)^3$, so every modulus $d = o(r)$ loses its phase error without a full cycle modulo d and without any primorial.**

A remark on the case $H = T$. There the Fourier multiplier of the two-stage weight is $|D_N(h)|^4/N^3 = \Delta_N(h/d)^2/N$ — the square of the Fejér kernel, which up to normalisation is the **Jackson kernel** of approximation theory. So at equal parameters the construction is a Jackson-type smoothing on the Fourier side. The spatial weight is a convolution of two triangles, not a Jackson kernel; the two should not be confused.

The exponent is sharp, and the construction has a name. Expanding $E_{d,b}$ over the non-zero frequencies and keeping one Fourier coefficient gives a lower bound for $\max_b|E_{d,b}|$. At $H = T = (d-1)/2$ one has $|D_H(1)| \asymp d$, so that lower bound and $d^3/(HT^2)$ are both of order 1, and no smaller power of d can hold uniformly in H and T. **This settles the exponent 3, not the sharpness of the constant** 1/45**.** Measured at that choice, $\max_b|E_{d,b}|$ against the bound is 0.225/0.237, 0.191/0.203 and 0.181/0.193 at $d = 11$, 23, 37 — a stable ratio near 0.94. The two-stage device is **iterated Fejér smoothing on** $\mathbb{Z}/d\mathbb{Z}$: the moving centre turns the time average into a second convolution on the same cyclic group, so the two Fejér multipliers multiply. Both the sharpness argument and the name are due to a respondent to a question by the author on MathOverflow; the numbers here are the author's check of them.

Proposition 3 makes the second of these usable across sectors, not just inside one: the moving centre samples consecutive sectors, whose errors are odd about a common centre, so the sliding average is a cancellation, not merely a smoothing.

Why the certificate survives smoothing. Since $w_H(u) = H^{-1}\sum_{h=1}^{H}\mathbf{1}_{\{|u|<h\}}$ and $q_T \ge 0$, a smoothed Bonferroni value is a positive average of sharp ones. So if the smoothed $\mathcal{L}_k$ exceeds 2, some sharp window in the family has $L_k > 2$, and by Proposition 1 that window holds at least three open cells — in the same sector, so Theorem 1 and Corollary 1 still apply. **Smoothing changes the estimate, not the target.**

What the smoothing buys, measured. Proposition 4 for the single tent and Proposition 5 for the double one, with the fixed choices $H = L/4$ and $H = T = L/8$ centred at the sector midpoint:

M	sharp L_7	tent L_7	double tent L_7	sharp L_9	tent L_9	double tent L_9
10,005	-93	-75.0	-43.2	502	120.6	59.4
20,001	-1,236	-286.0	-145.6	779	216.4	110.5
50,001	-7,494	-1,653.5	-791.2	1,567	395.7	199.2

The gain on L_7 is a factor of about nine at $M = 50{,}001$. **Still it is not enough: the sign does not change, and the order needed stays where Proposition 1 put it.** The kernels are the classical ones and the exponent is sharp, so this is not a failure of the chosen smoothing — it is as far as smoothing of this kind goes.

One further route, closed by an implication. Instead of smoothing line by line, one may smooth the survivor mask itself: with $A \subseteq \mathbb{Z}/Q\mathbb{Z}$ the cells surviving every line up to P, ask for

$$\Big|\sum_t q_T(t)\sum_u w_H(u)\ \mathbf{1}_A(c_t+u) - H\frac{|A|}{Q}\Big| < H\frac{|A|}{Q}$$

for every centre c, with H and T polynomial in P, not in the primorial Q. The primorial does drop out — the least $H = T$ that works is 2, 3, 5, 7, 10 at $P = 7, 11, 13, 17, 19$, against $Q = 35$ up to 1,616,615 — but not through a new mechanism:

P	Q	least $H = T$	largest gap g in A
7	35	2	5
11	385	3	7
13	5,005	5	11
17	85,085	7	18
19	1,616,615	10	25

The least admissible half-width is governed by that gap, and in one direction the link is a proof. The double tent is the convolution of two triangles, so its support is $4H - 3$ points, not $2H$. If a run of $4H - 3$ consecutive cells misses A entirely, its centre gives a smoothed count of exactly 0, breaking the strict inequality. Hence

$$4H - 3 \ \geq\ g, \qquad \text{that is} \qquad H \ \geq\ \Big\lceil \frac{g+3}{4} \Big\rceil,$$

which reads $2, 3, 4, 6, 7$ against the measured $2, 3, 5, 7, 10$ — satisfied in every row and attained in the first two. The converse fails, and the table shows why: the requirement is not just a positive count but a deviation below the mean, which is strictly stronger. So the measured H runs above the bound, and the ratio $4H - 3$ to g is not constant (5:5, 9:7, 17:11, 25:18, 37:25).

The proved direction is the one the route needs. The inequality with H polynomial in P forces g polynomial in P, and g is Jacobsthal's function on this residue system — open for over fifty years, with its best bound from the linear sieve, not Fourier analysis. The route is recorded as closed: a restatement, not a reduction. It is also weaker than what §2.1 needs, which is three survivors at a prescribed centre, not one at some centre.

Our reading of why is a matter of where the mass sits. Proposition 5 controls a modulus d only while $d \ll r$. Splitting each S_i by whether the product $d = p_1 \cdots p_i$ of its subset falls below r:

M	r	S_1	S_2	S_3	S_4	S_5–S_7
20,001	3,333	88.5%	40.1%	5.6%	0%	0%
50,001	8,333	90.0%	44.7%	8.7%	0.2%	0%

Smoothing reaches almost all of S_1 and none of S_5 and beyond. The arithmetic is immediate: for $p_1 \cdots p_i < r$ the primes must average $r^{1/i}$, which at $i = 4$ and $r = 8{,}333$ leaves only four subsets — $\{5, 7, 11, 13\}$, $\{5, 7, 11, 17\}$, $\{5, 7, 11, 19\}$ and $\{5, 7, 13, 17\}$ — and at $i = 5$ none. S_1 is the term that died at $M = 21$, and the terms that decide the sign of L_7 are S_4 through S_7, so **the smoothing controls exactly the part that never needed controlling.** On that reading we record a closed route: the boundary error does not seem to be what makes the sign negative, and since the moving kernel's exponent is sharp we do not expect another window shape to change it.

No progress toward the twin-prime conjecture is claimed, and no new bound. Priority is not claimed for any result.

References

The companion papers of this set are cited as [P1] to [P12], and the numbered entries below are the external works. The two kinds never share a number: a bracket with a P is a companion paper, a bare number is a reference in the list below. This paper imports only the definitions and statements of the companion papers, never their proofs.

1. G. Coppola and M. Laporta, *Sieve functions in arithmetic bands*, Hardy–Ramanujan J. **39** (2016), 21–37; arXiv:1503.07502.
2. H. G. Diamond and H. Halberstam, *A higher-dimensional sieve method*, Cambridge Tracts in Mathematics **177**, Cambridge University Press, 2008.
3. S. B. Ekhad, appendix to *Human and automated approaches for finite trigonometric sums*, arXiv:2204.08228 — Proposition Ton_2 and its higher analogues.
4. J. M. Flagg, *Primes between squares — commentary on Appendix 8 of Laws of Form*, arXiv:2511.05603 (2025) — cited for Lemma 1A only.
5. J. Grieshop, *Explicit constants for Fejér-type smoothing on finite cyclic groups*, arXiv:2512.03085. — *the closest treatment of the same kernel on the same group; its bound is of a different shape, and its Remark 5.4 sets aside the boundary error that (B.4.1) evaluates.*
6. H. Halberstam and H.-E. Richert, *Sieve Methods*, Academic Press, 1974.
7. H. Iwaniec, *Almost-primes represented by quadratic polynomials*, Invent. Math. **47** (1978), 171–188.
8. R. J. Lemke Oliver, *Almost-primes represented by quadratic polynomials*, Acta Arith. **151** (2012), 241–261.
9. H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory III*, in preparation. — *Exercise 24.2.1.1(d), the pointwise Fejér estimate cited in Appendix B.4; the exercise number may move, the volume being unpublished.*
10. G. Morpurgo, *On a stricter twin primes conjecture, and on the Polignac's conjecture in general*, arXiv:2210.15487 (2022; revised 2023). — *A preprint, not peer reviewed; a heuristic prediction*

with numerical comparison, containing no theorems. Cited as contemporaneous independent work using the same coordinates.

11. T. T. K. Nguyen, *Finite-window noncovering on primorial wheels: higher-order CRT bounds and shift correlations*, Preprints.org (2026), doi:10.20944/preprints202608.1299.v1. — *A preprint, not peer reviewed; cited as contemporaneous independent work reaching the same finite-window diagnosis from the Goldbach side.*
12. A. Schinzel and W. Sierpiński, *Sur certaines hypothèses concernant les nombres premiers*, Acta Arith. **4** (1958), 185–208; erratum, ibid. **5** (1959), 259.
13. M. Bordignon and E. S. Lee, *Explicit upper bounds for the number of primes simultaneously representable by any set of irreducible polynomials*, arXiv:2211.11012. — *equation (2) there states the upper bound used for Corollary 1e', citing [6, Thm. 5.3]; the entry was added for that citation and its proof is not re-derived here.*